CHAPTER 4

Upper Limb

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CONTENTS

RADIOGRAPHIC ANATOMY

Upper Limb

The bones of the upper limb can be divided into four main groups: (1) **hand and wrist,** (2) **forearm,** (3) **arm** (humerus), and (4) **shoulder girdle** (Fig. 4.1). The first two groups are discussed in this chapter. The important wrist and elbow joints are included; the shoulder joint and proximal humerus are discussed in Chapter 5.

The shape and structure of each of the bones and articulations, or joints, of the upper limb must be thoroughly understood by technologists so that each part can be identified and demonstrated on radiographs.

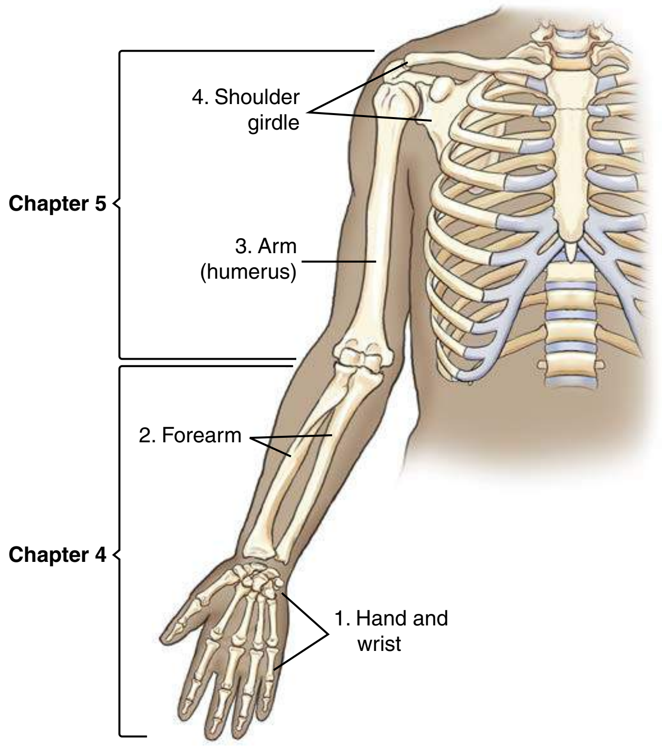

Fig. 4.1 Right upper limb (anterior view).

HAND AND WRIST

The 27 bones in each hand and wrist are divided into the following three groups (Fig. 4.2):

Phalanges (fingers and thumb)	14
Metacarpals (palm)	5
Carpals (wrist)	8
Total	27

The most distal bones of the hand are the **phalanges** *(fa-lan′-jez),* which constitute the digits (fingers and thumb). The second group of bones is the **metacarpals** *(met″-ah-kar′-palz)*; these bones make up the palm of each hand. The third group of bones, the **carpals** *(kar′-palz),* consists of the bones of the wrist.

Phalanges: Fingers and Thumb (Digits)

Each finger and thumb is called a *digit,* and each digit consists of two or three separate small bones called *phalanges* (singular, **phalanx** *[fa′-lanks]*). The digits are numbered, starting with the thumb as 1 and ending with the little finger as 5.

Each of the four fingers (digits 2, 3, 4, and 5) is composed of three phalanges—**proximal, middle,** and **distal.** The thumb, or first digit, has two phalanges—**proximal** and **distal.**

Each phalanx consists of three parts: a distal rounded **head,** a **body** (shaft), and an expanded **base,** similar to that of the metacarpals.

Metacarpals (Palm)

The second group of bones of the hand, which make up the palm, consists of the five **metacarpals.** These bones are numbered the same way as the digits are, with the first metacarpal being on the thumb, or lateral, side when the hand is in the anatomic position.

4

Each metacarpal is composed of three parts, similar to the phalanges. Distally, the rounded portion is the **head.** The **body** (shaft) is the long curved portion; the anterior part is concave in shape, and the posterior, or dorsal, portion is convex. The **base** is the expanded proximal end, which articulates with associated carpals.

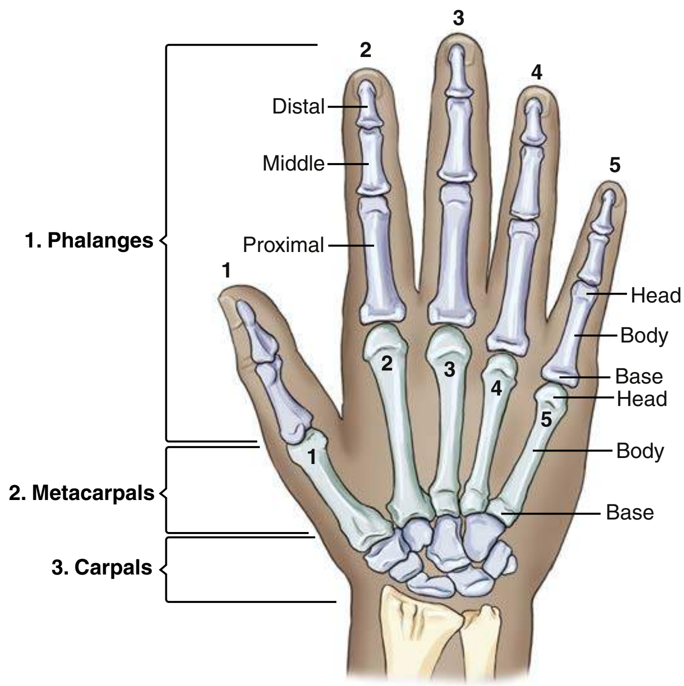

Fig. 4.2 Right hand and wrist (posterior view).

JOINTS OF THE HAND

The joints, or articulations, between the individual bones of the upper limb are important in radiology because small chip fractures may occur near the joint spaces. Therefore, accurate identification of all joints of the phalanges and metacarpals of the hand is required (Fig. 4.3).

Thumb (First Digit)

The thumb has only two phalanges, so the joint between them is called the **interphalangeal (IP) joint.** The joint between the first metacarpal and the proximal phalanx of the thumb is called the **first metacarpophalangeal (MCP) joint.** The name of this joint consists of the names of the two bones that make up this joint. The proximal bone is named first, followed by the distal bone.

For radiographic purposes, the first metacarpal is considered part of the thumb and must be included in its entirety in a radiograph of the thumb—**from the distal phalanx to the base of the first metacarpal.** This inclusion is not the case with the fingers, which for positioning purposes include only the three phalanges—distal, middle, and proximal.

Fingers (Second through Fifth Digits)

Each of the second through fifth digits has three phalanges, and they have three joints each. Starting from the most distal portion of each digit, the joints are the **distal interphalangeal (DIP) joint,** followed by the **proximal interphalangeal (PIP) joint,** and, most proximally, the **MCP joint.**

MPJ and CMC Joints

The metacarpals articulate with the phalanges at their distal ends and are called **metacarpophalangeal (MCP) joints.** At the proximal end, the metacarpals articulate with the respective carpals and are called **carpometacarpal (CMC) joints.** The five metacarpals articulate with specific carpals as follows:

- First metacarpal with trapezium
- Second metacarpal with trapezoid
- Third metacarpal with capitate
- Fourth and fifth metacarpal with hamate

REVIEW EXERCISE WITH RADIOGRAPHS

In identifying the joints and phalanges of the hand, the specific digit and hand must be included in the descriptions. A PA radiograph of the hand (Fig. 4.4) shows the phalanges and metacarpals and the joints described previously. A good review exercise is to identify each part labeled A through R on Fig. 4.4 (cover up the answers listed next). Then check your answers against the following list:

A. First carpometacarpal joint of right hand
B. First metacarpal of right hand
C. First metacarpophalangeal joint of right hand
D. Proximal phalanx of first digit (or thumb) of right hand
E. Interphalangeal joint of first digit (or thumb) of right hand
F. Distal phalanx of first digit (or thumb) of right hand
G. Second metacarpophalangeal joint of right hand
H. Proximal phalanx of second digit of right hand
I. Proximal interphalangeal joint of second digit of right hand
J. Middle phalanx of second digit of right hand
K. Distal interphalangeal joint of second digit of right hand
L. Distal phalanx of second digit of right hand
M. Middle phalanx of fourth digit of right hand
N. Distal interphalangeal joint of fifth digit of right hand
O. Proximal phalanx of third digit of right hand
P. Fifth metacarpophalangeal joint of right hand
Q. Fourth metacarpal of right hand
R. Fifth carpometacarpal joint of right hand

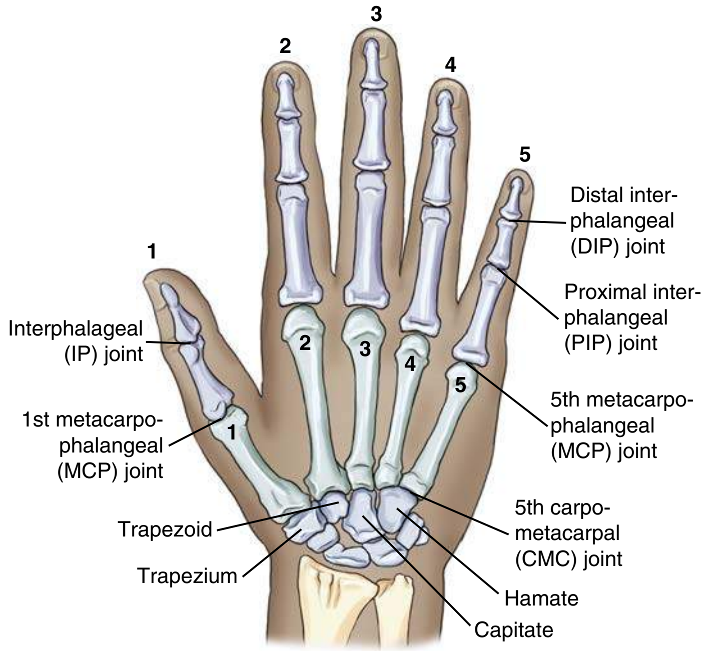

Fig. 4.3 Joints of right hand and wrist.

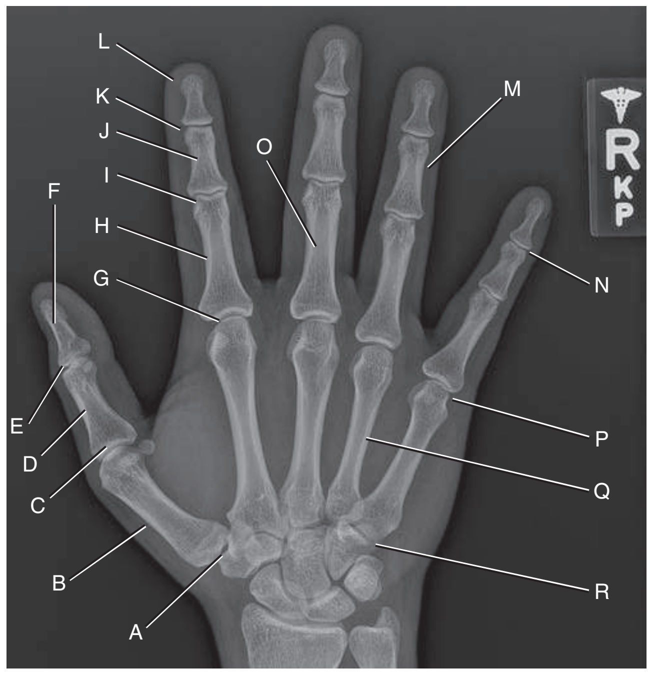

Fig. 4.4 PA radiograph of right hand.

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CARPALS (WRIST)

The third group of bones of the hand and wrist are the **carpals,** the bones of the wrist. Learning the names of the eight carpals is easiest when they are divided into two rows of four each (Fig. 4.5).

Proximal Row

Beginning on the lateral, or thumb, side is the **scaphoid** *(skaf′-oyd),* sometimes referred to as the *navicular.* One of the tarsal bones of the foot also is sometimes called the *navicular* or *scaphoid.* However, the correct term for the tarsal bone of the **foot** is the **navicular,** and the correct term for the carpal bone of the **wrist** is the **scaphoid.**

The scaphoid, a boat-shaped bone, is the largest bone in the proximal row and **articulates with the radius proximally.** Its location and articulation with the forearm make it important radiographically because it is the **most frequently fractured carpal bone.**

The **lunate** (moon shaped) is the second carpal in the proximal row; it **articulates with the radius.** It is distinguished by the deep concavity on its distal surface, where it articulates with the capitate of the distal row of carpals (best seen on anterior view; Fig. 4.6).

The **third** carpal is the **triquetrum** *(tri-kwe′-trum),* which has three articular surfaces and is distinguished by its pyramidal shape and anterior articulation with the small pisiform.

The **pisiform** *(pi′-si-form)* (pea shaped), the smallest of the carpal bones, is located anterior to the triquetrum and is most evident in the carpal canal or tangential projection (Fig. 4.7).

Distal Row

The second, more distal row of four carpals articulates with the five metacarpal bones. Starting again on the lateral, or thumb, side is the **trapezium** *(trah-pe′-ze-um),* a four-sided, irregularly shaped bone that is located medial and distal to the scaphoid and proximal to the first metacarpal. The wedge-shaped **trapezoid** *(trap′-e-zoyd),* also four sided, is the smallest bone in the distal row. This bone is followed by the largest of the carpal bones, the **capitate** *(kap′-i-tate)* (capitate means "large bone"). It is identified by its large rounded head that fits proximally into a concavity formed by the scaphoid and lunate bones.

The last carpal in the distal row on the medial aspect is the **hamate** *(ham′-ate),* which is easily distinguished by the hooklike process called the **hamulus** *(ham′-u-lus),* or hamular process, which projects from its palmar surface (see Fig. 4.7).

Carpal Sulcus (Canal or Tangential Projection)

Fig. 4.7 is a drawing of the carpals as they would appear in a tangential projection down the wrist and arm from the palm or volar side of a hyperextended wrist. This view demonstrates the carpal sulcus formed by the concave anterior or palmar aspect of the carpals. The anteriorly located pisiform and the hamulus process of the hamate are visualized best on this view. This concave area or groove is called the *carpal sulcus (carpal tunnel* or *canal),* through which major nerves and tendons pass.

The term *hamate* means hooked, which describes the shape of the hamate in the illustration. The trapezium and its relationships to the thumb and trapezoid are well demonstrated.

Summary Chart of Carpal Terminology

The terms listed in Table 4.1 are used throughout this text. The names of these eight carpals may be remembered more easily with the use of a mnemonic (in which a sentence or phrase is formed by using the first letter of each carpal). Two mnemonic samples are provided in Table 4.1.

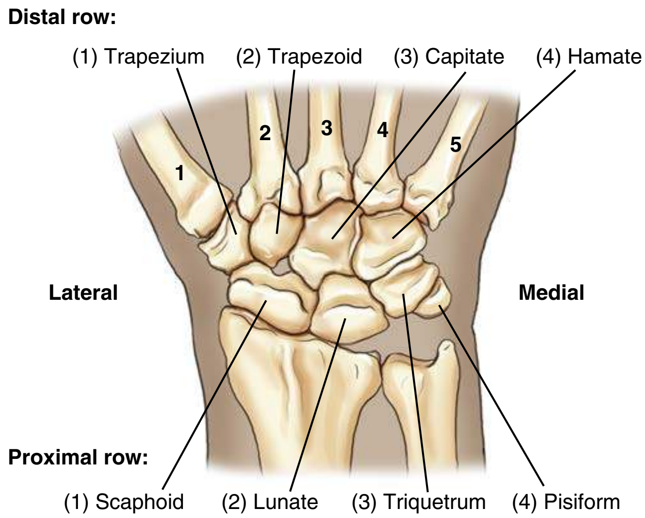

Fig. 4.5 Right carpals (dorsal or posterior view).

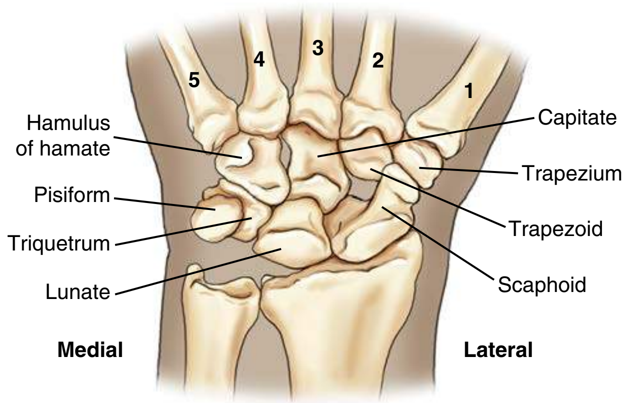

Fig. 4.6 Right carpals (palmar or anterior view).

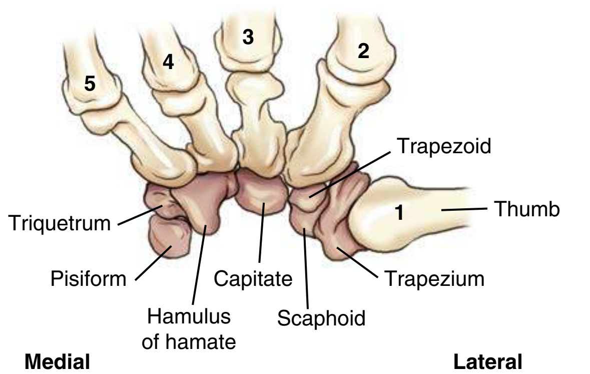

Fig. 4.7 Carpal sulcus (carpal canal or tangential projection).

TABLE 4.1 MNEMONIC FOR THE CARPALS

MNEMONIC			CARPAL
Send		Steve	Scaphoid
Letter		Left	Lunate
To		The	Triquetrum
Peter	Or	Party	Pisiform
To		To	Trapezium
Tell 'em (to)		Take	Trapezoid
Come		Carol	Capitate
Home		Home	Hamate

REVIEW EXERCISE WITH RADIOGRAPHS

Five projections for the wrist are shown in Figs. 4.8 through 4.12. A good review exercise is to identify each carpal bone as labeled (first cover the answers that follow). Check your answers against the following list.

In the lateral position (see Fig. 4.12), the trapezium (E) and the scaphoid (A) are located more anteriorly. Also, the ulnar deviation projection (see Fig. 4.10) best demonstrates the scaphoid without the foreshortening and overlapping seen on the posteroanterior (PA) (see Fig. 4.8).

The radial deviation projection (see Fig. 4.9) best demonstrates the interspaces and the carpals on the ulnar (lateral) side of the wrist-hamate (H), triquetrum (C), pisiform (D), and lunate (B). The outline of the end-on view of the hamulus process of the hamate (h) also can be seen on this radial deviation radiograph. The hamulus process also is demonstrated well on the carpal canal projection in Fig. 4.11, as is the pisiform (D), which is projected anteriorly and is seen in its entirety. Answers are as follows:

A. Scaphoid
B. Lunate
C. Triquetrum
D. Pisiform
E. Trapezium
F. Trapezoid
G. Capitate
H. Hamate
h. Hamulus (hamular process of hamate)

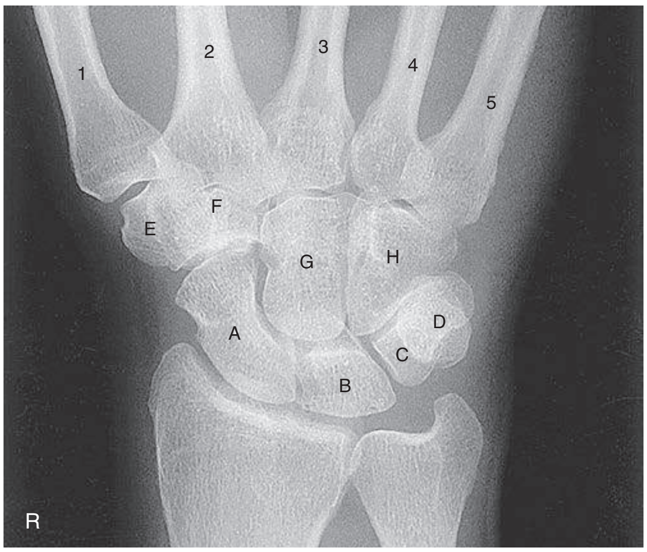

Fig. 4.8 PA wrist.

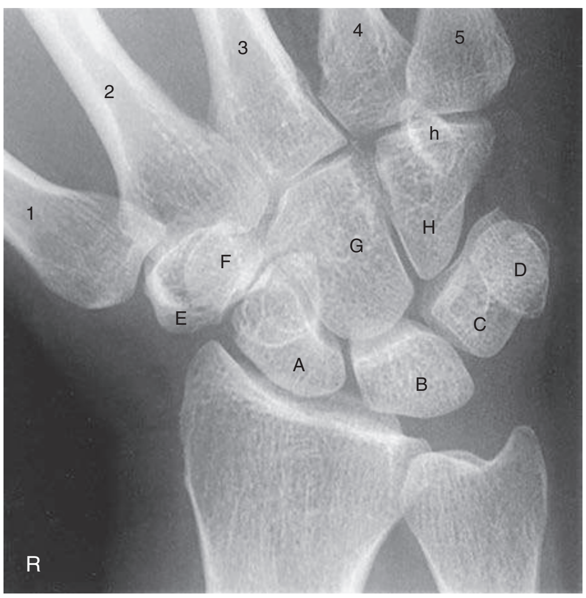

Fig. 4.9 Radial deviation.

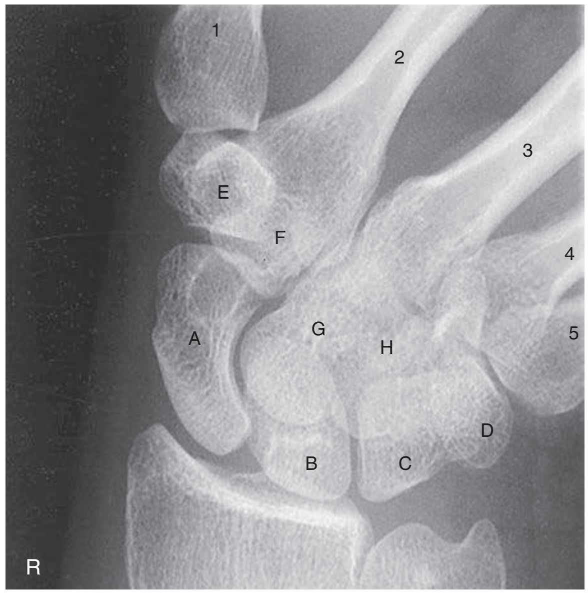

Fig. 4.10 Ulnar deviation (for scaphoid).

Fig. 4.11 Carpal canal. The scaphoid (A) is partially superimposed with the trapezium (E) and the trapezoid (F) on this projection.

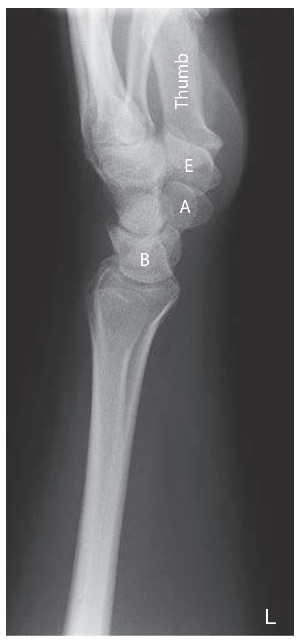

Fig. 4.12 Lateral wrist.

FOREARM—RADIUS AND ULNA

The second group of upper limb bones consists of the bones of the forearm—the radius on the lateral or thumb side and the ulna on the medial side (Fig. 4.13). The radius and ulna articulate with each other at the proximal radioulnar joint and at the distal radioulnar joint, as shown in Fig. 4.14. These two joints allow for the rotational movement of the wrist and hand, as described later in this chapter.

Radius and Ulna

Small conical projections, called **styloid processes,** are located at the extreme distal ends of both the radius and the ulna (see Fig. 4.14). The radial styloid process can be palpated on the thumb side of the wrist joint. The radial styloid process extends more distally than the ulnar styloid process.

The **ulnar notch** is a small depression on the medial aspect of the distal radius. The head of the ulna fits into the ulnar notch to form the distal radioulnar joint.

The **head of the ulna** is located near the wrist at the **distal** end of the ulna. When the hand is pronated, the ulnar head and the styloid process are easily felt and seen on the "little finger" side of the distal forearm.

The **head of the radius** is located at the **proximal** end of the radius near the elbow joint. The long midportion of both the radius and the ulna is called the **body** *(shaft).*

The radius, the shorter of the two bones of the forearm, is the only one of the two that is directly involved in the wrist joint. During the act of pronation, the radius is the bone that rotates around the more stationary ulna.

The proximal radius shows the round, disklike **head** and the **neck** of the radius as a tapered constricted area directly below the head. The rough oval process on the medial and anterior side of the radius, just distal to the neck, is the **radial tuberosity.**

Proximal Ulna The ulna, the longer of the two bones of the forearm, is primarily involved in the formation of the elbow joint. The two beaklike processes of the proximal ulna are called the **olecranon** and the **coronoid processes** (Figs. 4.14 and 4.15). The olecranon process can be palpated easily on the posterior aspect of the elbow joint.

The medial margin of the coronoid process opposite the radial notch *(lateral)* is commonly referred to as the **coronoid tubercle** (see Fig. 4.14 and anteroposterior [AP] elbow radiograph in Fig 4.19).

The large concave depression, or notch, that articulates with the distal humerus is the **trochlear** *(trok′-le-ar)* **notch** (semilunar notch). The small, shallow depression located on the lateral aspect of the proximal ulna is the **radial** *(ra′-de-al)* **notch.** The head of the radius articulates with the ulna at the radial notch, forming the proximal radioulnar joint. This joint, or articulation, is the proximal radioulnar joint that combines with the distal radioulnar joint to allow rotation of the forearm during pronation. During the act of pronation, the radius crosses over the ulna near the upper third of the forearm (see Fig. 4.25).

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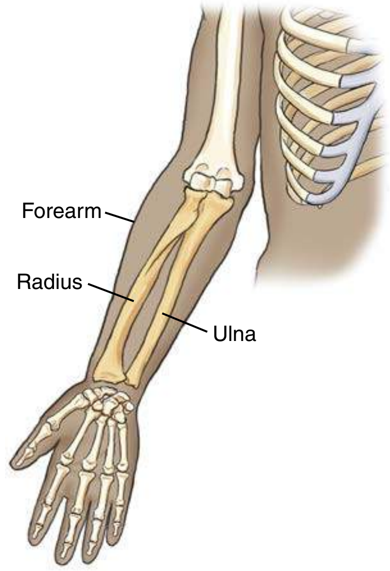

Fig. 4.13 Right upper limb (anterior view).

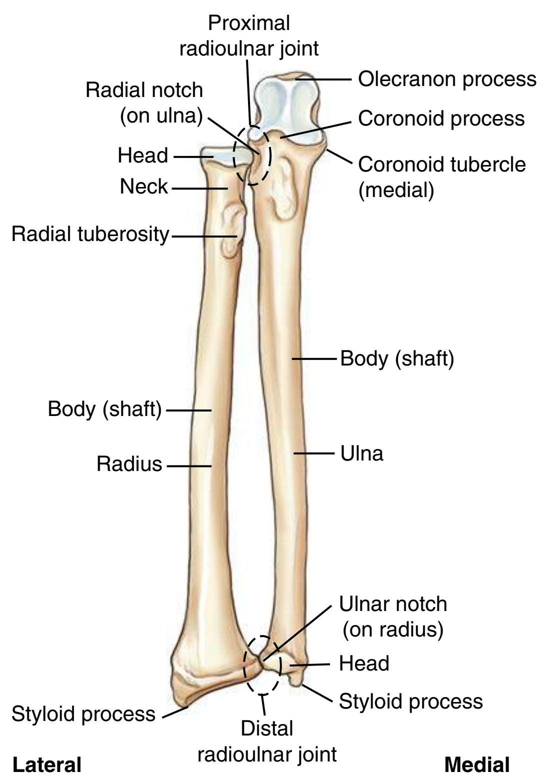

Fig. 4.14 Right radius and ulna (anterior view).

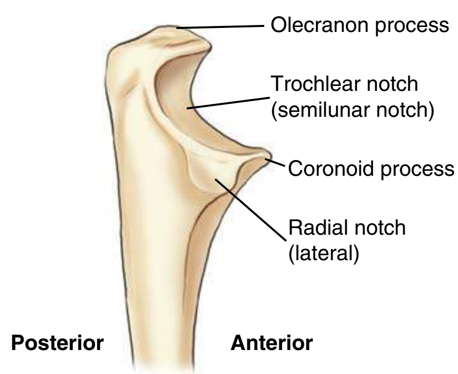

Fig. 4.15 Proximal ulna (lateral view).

DISTAL HUMERUS

Specific structures of the proximal humerus are discussed along with the shoulder girdle in Chapter 5. However, the anatomy of the midhumerus and distal humerus is included in this chapter as part of the elbow joint.

The **body** (shaft) of the humerus is the long center section, and the expanded distal end of the humerus is the **humeral condyle.** The articular portion of the humeral condyle is divided into two parts: the **trochlea** *(troke′-le-ah) (medial condyle)* and the **capitulum** *(kah-pit′-u-lum) (lateral condyle).*

The **trochlea** (meaning "pulley") is shaped like a pulley or spool; it has two rimlike outer margins and a smooth depressed center portion called the **trochlear sulcus,** or *groove.* This depression of the trochlea, which begins anteriorly and continues inferiorly and posteriorly, appears circular on a lateral end-on view; on a lateral elbow radiograph, it appears as a less dense (more radiolucent) area (see Figs. 4.17 and 4.20). The **trochlea** is located more medially and articulates with the **ulna.**

The **capitulum,** meaning "little head," is located on the lateral aspect and articulates with the head of the **radius.** (A memory aid is to associate the **cap**itulum ["cap"] with the "head" of the radius.) In earlier literature, the capitulum was called the **capitellum** *(kap″-i-tel′-um).*

The articular surface that makes up the rounded articular margin of the capitulum is just slightly smaller than that of the trochlea (Fig. 4.18). This structure becomes significant in the evaluation for a true lateral position of the elbow, as does the direct superimposition of the two **epicondyles** *(ep″-e-kon′-dylz).*

The **lateral epicondyle** is the small projection on the lateral aspect of the distal humerus above the capitulum. The **medial epicondyle** is larger and more prominent than the lateral epicondyle and is located on the medial edge of the distal humerus. In a true lateral position, the directly superimposed epicondyles (which are difficult to recognize) are seen as proximal to the circular appearance of the trochlear sulcus (see Fig. 4.17).

The distal humerus has specific **depressions** on both anterior and posterior surfaces. The two shallow **anterior depressions** are the **coronoid fossa** and the **radial fossa** (see Figs. 4.16 and 4.17. As the elbow is completely flexed, the coronoid process and the radial head are received by these respective fossae, as the names indicate.

The deep **posterior depression** of the distal humerus is the **olecranon fossa** (not specifically shown on these illustrations). The olecranon process of the ulna fits into this depression when the arm is fully extended. Soft tissue detail as depicted by specific fat pads located within the deep olecranon fossa is important in trauma diagnosis of the elbow joint.

The lateral view of the elbow (see Fig. 4.17) clearly shows specific parts of the proximal radius and ulna. The **head** and **neck** of the radius are well demonstrated, as are the **radial tuberosity** (partially seen on the proximal radius) and the large concave **trochlear (semilunar) notch.**

TRUE LATERAL ELBOW

Specific positions, such as an **accurate lateral** with **90° flexion,** along with possible associated visualization of fat pads, are essential for evaluation of joint pathology of the elbow.

A good criterion by which to evaluate a true lateral position of the elbow when it is flexed 90° is the appearance of the three concentric arcs, as labeled in Fig. 4.18. The first and smallest arc is the **trochlear sulcus.** The second, intermediate arc appears double lined as the outer ridges or rounded edges of the **capitulum** and **trochlea.**[1] (The smaller of the double-lined ridges is the capitulum; the larger is the medial ridge of the trochlea.) The **trochlear notch of the ulna** appears as a third arc of a true lateral elbow. If the elbow is rotated even slightly from a **true** lateral, the arcs do not appear symmetrically aligned in this way, and the elbow joint space is not as open.

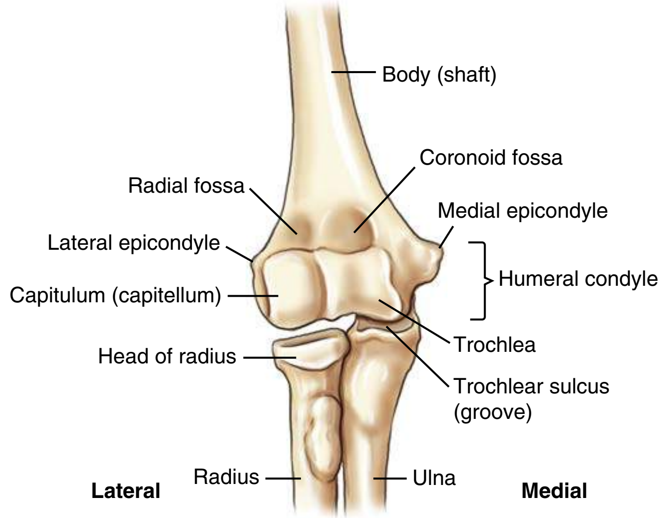

Fig. 4.16 Distal humerus (anterior view).

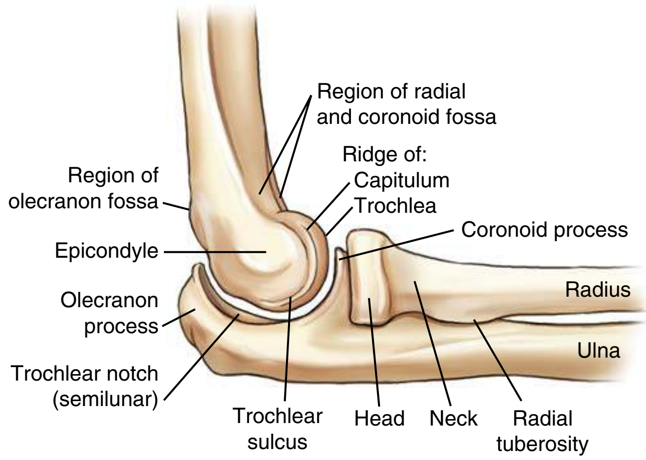

Fig. 4.17 Lateral elbow.

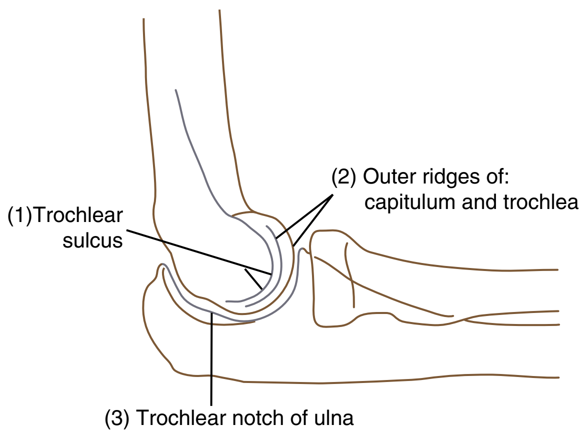

Fig. 4.18 True lateral elbow—three concentric arcs.

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REVIEW EXERCISE WITH RADIOGRAPHS

The AP and lateral radiographs of the elbow provide a review of anatomy and demonstrate the three concentric arcs as evidence of a true lateral position (Figs. 4.19 and 4.20). Answers to the labels are as follows:

A. Medial epicondyle
B. Trochlea (medial aspect)
C. Coronoid tubercle
D. Radial head
E. Capitulum
F. Lateral epicondyle
G. Superimposed epicondyles of humerus
H. Olecranon process
I. Trochlear sulcus
J. Trochlear notch
K. Double outer ridges of capitulum and trochlea (capitulum being the smaller of the two areas and trochlea the larger)
L. Coronoid process of ulna
M. Radial head
N. Radial neck

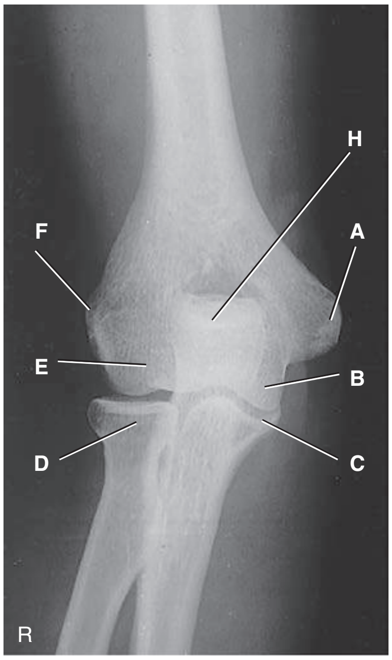

Fig. 4.19 AP elbow.

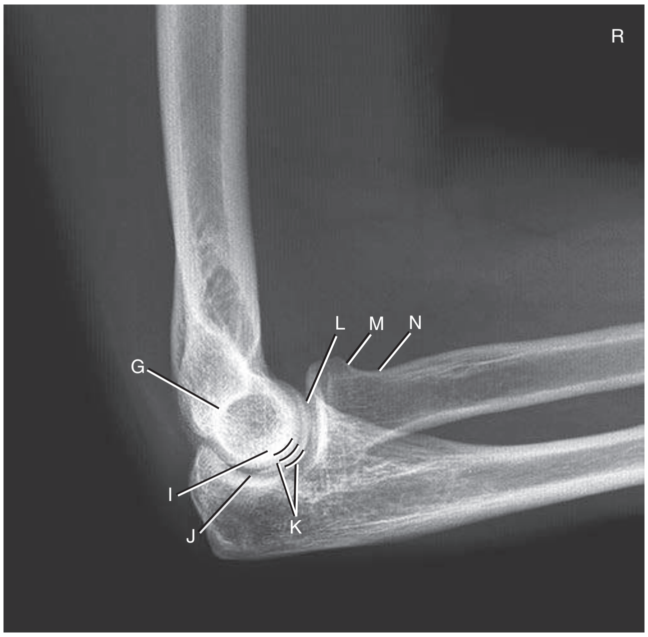

Fig. 4.20 Lateral elbow.

CLASSIFICATION OF JOINTS

Table 4.2 provides a summary of hand, wrist, forearm, and elbow joints. Refer back to Chapter 1 for a general description of joints or articulations, along with the various classifications and movement types. These classifications are reviewed and described here more specifically for each joint of the hand, wrist, forearm, and elbow.

All joints of the upper limb as described in this chapter are classified as **synovial** and are freely movable, or **diarthrodial.** Only the movement types differ.

Hand and Wrist

Interphalangeal Joints Beginning distally with the phalanges, all IP joints are **ginglymus,** or hinge-type, joints with movement in two directions only—**flexion** and **extension** (Fig. 4.21). This movement occurs in one plane only, around the transverse axis. This includes the single IP joint of the thumb (first digit) and the distal and proximal IP joints of the fingers (second to fifth digits).

Metacarpophalangeal Joints The second to fifth MCP joints are ellipsoidal (condyloid)-type joints that allow movement in four directions—flexion, extension, abduction, and adduction. Circumduction movement, which also occurs at these joints, is conelike sequential movement in these four directions.

The first MCP joint (thumb) also is generally classified as an ellipsoidal (condyloid) joint, although it has very limited abduction and adduction movements because of the wider and less-rounded head of the first metacarpal (see Fig. 4.21).

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Carpometacarpal Joints The first CMC joint of the thumb is a **saddle** (sellar)-type joint. This joint best demonstrates the shape and movement of a saddle joint, which allows a great range of movement, including flexion, extension, abduction, adduction, circumduction, opposition, and some degree of rotation.

The second through fifth CMC joints are **plane** (gliding)-type joints, which allow the least amount of movement of the synovial class joints. The joint surfaces are flat or slightly curved, with movement limited by a tight fibrous capsule.

Intercarpal Joints The intercarpal joints between the various carpals have only a **plane (gliding)** movement.

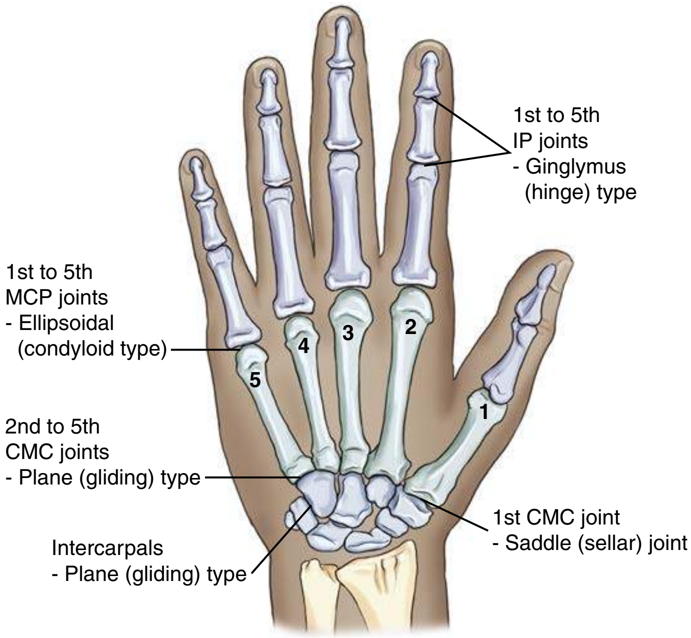


Fig. 4.21 Joints of left hand and wrist (posterior view).

Wrist Joint

The wrist joint is an **ellipsoidal** (condyloid)-type joint and is the most freely movable, or **diarthrodial,** of the **synovial classification.** Of the two bones of the forearm, only the radius articulates directly with two carpal bones—the **scaphoid** and the **lunate.** This wrist joint is called the **radiocarpal joint.**

The **triquetral** bone is also part of the wrist joint in that it is opposite the **articular disk.** The articular disk is part of the total wrist articulation, including a joint between the distal radius and ulna of the forearm—the **distal radioulnar joint.**

The articular surface of the distal radius along with the total articular disk forms a smooth, concave-shaped articulation with the three carpals to form the complete wrist joint.

The total wrist joint is enclosed by an articular synovial capsule that is strengthened by ligaments that allow movement in four directions, plus circumduction.

The synovial membrane lines the synovial capsule and the four wrist ligaments as they pass through the capsule, in addition to lining the distal end of the radius and the articular surfaces of adjoining carpal bones.

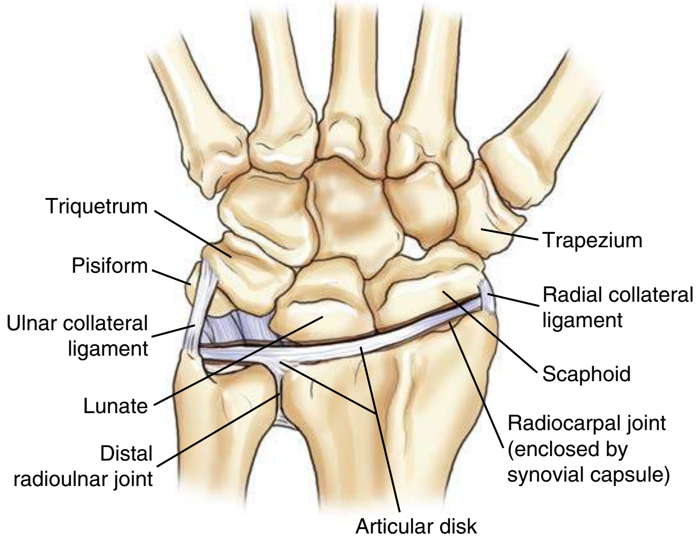


Fig. 4.22 Left wrist joint with articular disk (posterior view).

Wrist Ligaments

The wrist has numerous important ligaments that stabilize the wrist joint. Two of these are shown in the drawing in Fig. 4.22. The **ulnar collateral ligament** is attached to the styloid process of the ulna and fans out to attach to the triquetrum and the pisiform. The **radial collateral ligament** extends from the styloid process of the radius primarily to the lateral side of the scaphoid (scaphoid tubercle), but it also has attachments to the trapezium.

Five additional ligaments not shown in this drawing are crucial to the stability of the wrist joint and often are damaged during trauma. These five ligaments are commonly imaged with conventional arthrography or magnetic resonance imaging (MRI):

- Dorsal radiocarpal ligament
- Palmar radiocarpal ligament
- Triangular fibrocartilage complex (TFCC)
- Scapholunate ligament
- Lunotriquetral ligament

Elbow Joint

The elbow joint is also of the **synovial classification** and is freely movable, or **diarthrodial.** The elbow joint generally is considered a **ginglymus** (hinge)-type joint with flexion and extension movements between the humerus and the ulna and radius. However, the complete elbow joint includes three joints enclosed in one articular capsule. In addition to the hinge joints between the humerus and ulna and the humerus and radius, the **proximal radioulnar joint** (**pivot** or trochoidal type) is considered part of the elbow joint (Fig. 4.23).

The importance of accurate lateral positioning of the elbow for visualization of certain fat pads within the elbow joint is discussed later in this chapter.

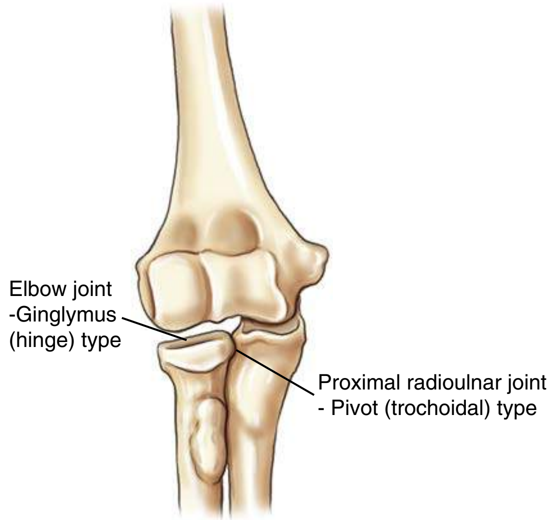


Fig. 4.23 Elbow joint.

TABLE 4.2 SUMMARY OF HAND, WRIST, FOREARM, AND ELBOW JOINTS

Classification	
Synovial (articular capsule containing synovial fluid)	
Mobility Type	
Diarthrodial (freely movable)	
Movement Type	
1. Interphalangeal joints	**Ginglymus** (hinge)
2. Metacarpophalangeal joints	**Ellipsoidal** (condyloid)
3. Carpometacarpal joints	
First digit (thumb)	**Saddle** (sellar)
Second to fifth digits	**Plane** (gliding)
4. Intercarpal joints	**Plane** (gliding)
5. Wrist (radiocarpal) joint	**Ellipsoidal** (condyloid)
6. Proximal: radioulnar	**Pivot** (trochoidal) joint
7. Elbow joint	
Humeroulnar and humeroradial	**Ginglymus** (hinge)

4

WRIST JOINT MOVEMENT TERMINOLOGY

Certain terminology involving movements of the wrist joint may be confusing, but these terms must be understood by technologists because special projections of the wrist are described by these movements. These terms were described in Chapter 1 as turning or bending the hand and wrist from its natural position toward the side of the ulna for **ulnar deviation** and toward the radius for **radial deviation** (Fig. 4.24).

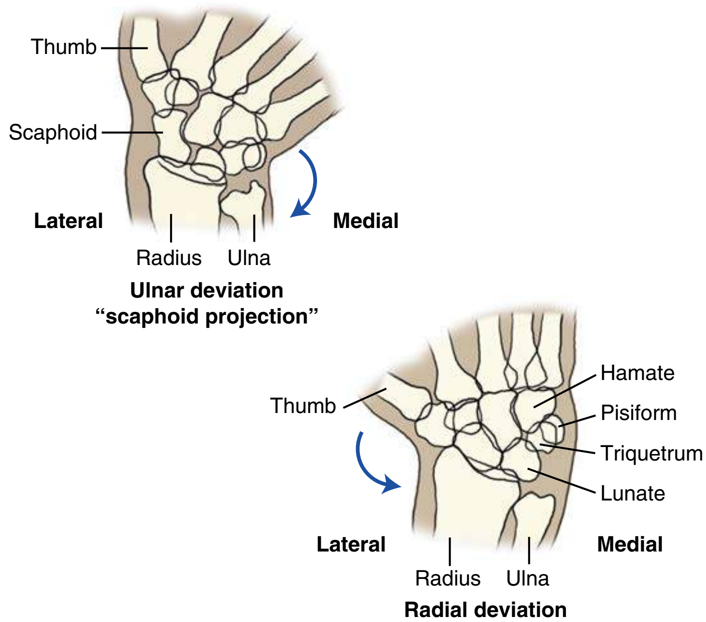


Fig. 4.24 Wrist movements.

Ulnar Deviation

The ulnar deviation movement of the wrist "opens up" and best demonstrates the carpals on the opposite side (the radial or lateral side) of the wrist—the scaphoid, trapezium, and trapezoid. Because the scaphoid is the most frequently fractured carpal bone, this ulnar deviation projection is commonly known as a *special scaphoid projection.*

Radial Deviation

A less frequent PA wrist projection involves the radial deviation movement that opens and best demonstrates the carpals on the opposite, or ulnar, side of the wrist—the hamate, pisiform, triquetrum, and lunate.

FOREARM ROTATIONAL MOVEMENTS

The radioulnar joints of the forearm also involve some special rotational movements that must be understood for accurate imaging of the forearm. For example, the forearm generally should not be radiographed in a pronated position (a PA projection), which may appear to be the most natural position for the forearm and hand. The forearm routinely should be radiographed in an AP projection with the hand supinated, or palm up (anatomic position). The reason becomes clear in studying the "cross-over" position of the radius and ulna when the hand is pronated (Fig. 4.25). This cross-over results from the unique pivot-type rotational movements of the forearm that involve both the proximal and the distal radioulnar joints.

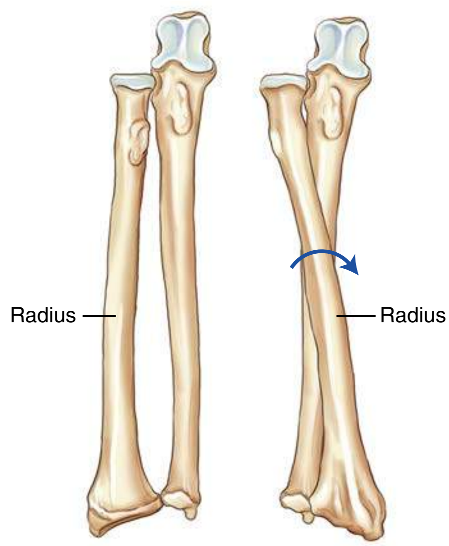


Fig. 4.25 Forearm rotational movements.

Summary To prevent superimposition of the radius and ulna that may result from these pivot-type rotational movements, the forearm is radiographed with the **hand supinated** for an **AP projection.**

ELBOW ROTATIONAL MOVEMENTS

The appearance of the proximal radius and ulna changes as the elbow and distal humerus are rotated or positioned obliquely either medially or laterally as shown on these radiographs. On the AP radiograph with no rotation, the proximal radius is superimposed only slightly by the ulna (Fig. 4.26).

The radius and ulna can be separated through lateral rotation (40-45 degrees) of the elbow, as shown in Fig. 4.27, whereas medial rotation (pronated hand) completely superimposes them, as can be seen in Fig. 4.28. This relationship is crucial in evaluation of AP projections of the elbow; **lateral rotation separates** and **medial rotation superimposes** the proximal radius and ulna.

Fig. 4.26 AP, **no rotation**—radius and ulna partially superimposed.

Fig. 4.27 AP, **lateral rotation**—separation of radius and ulna.

Fig. 4.28 AP, **medial rotation**—superimposed radius and ulna.

IMPORTANCE OF VISUALIZING FAT PADS

Radiographs of the upper and lower limbs are taken not only to evaluate for disease or trauma to bony structures but also to assess associated soft tissues, such as certain accumulations of fat called **fat pads, fat bands,** or **stripes.** In some cases, displacement of an adjoining fat pad or band may be the only indication of disease or significant injury or fracture within a joint region.

For diagnostic purposes, the most important fat pads or bands are those located around certain joints of the upper and lower limbs. These fat pads are extrasynovial (outside the synovial sac) but are located within the joint capsule. Therefore, any changes that occur within the capsule itself alter the normal position and shape of the fat pads. Most often, such changes result from fluid accumulation (effusion) within the joint, which indicates the presence of an injury involving the joint.

Radiolucent fat pads are seen as densities that are slightly more lucent than surrounding structures. Fat pads and their surrounding soft tissue are of only slightly different tissue density (brightness), making them difficult to visualize on radiographs. This visualization requires optimum exposure for visualization of these soft tissue structures.[2] (They generally are not visible on published radiographs without enhancement, as is shown on the illustrations on this page.)

Wrist Joint

The wrist joint includes two important fat stripes. First, a **scaphoid fat stripe** (A) is visualized on the PA (Fig. 4.29) and oblique (Fig. 4.30) projections. It is elongated and slightly convex in shape and is located between the radial collateral ligament and adjoining muscle tendons immediately lateral to the scaphoid. Absence or displacement of this fat stripe may be the only indicator of a fracture on the radial aspect of the wrist.

A second fat stripe is visualized on the lateral view of the wrist. This **pronator fat stripe** (B) is normally visualized approximately ¼ inch (1 cm) from the anterior surface of the radius (Fig. 4.31). Subtle fractures of the distal radius can be indicated by displacement or obliteration of the plane of this fat stripe.[2]

Elbow Joint

The three significant fat pads or stripes of the elbow are visualized only on the lateral projection. They are not seen on the AP because of their superimposition over bony structures. On the lateral projection, the **anterior fat pad** (Fig. 4.32C), which is formed by the superimposed coronoid and radial pads, is seen as a slightly radiolucent teardrop shape located just anterior to the distal humerus. Trauma or infection can cause the anterior fat pad to be elevated and more visible and distorted in shape. This is usually visible only on a true lateral elbow projection flexed 90°.

The **posterior fat pad** (see Fig. 4.32D) is located deep within the olecranon fossa and normally is **not visible** on a negative elbow examination. Visualization of this fat pad on a 90° flexed lateral elbow radiograph indicates that a change within the joint has caused its position to change, suggesting the presence of a joint pathologic process.

To ensure an accurate diagnosis, the elbow **must be flexed 90°** on the lateral view. If the elbow is extended beyond the 90° flexed position, the olecranon slides into the olecranon fossa, elevates the posterior fat pad, and causes it to appear. In this situation, the pad is visible whether the examination is negative or positive. Generally, visualization of the posterior fat pad is considered more reliable than visualization of the anterior fat pads.

The **supinator fat stripe** (see Fig. 4.32E) is a long, thin stripe just anterior to the proximal radius. It may indicate the diagnosis of radial head or neck fractures that are not obviously apparent.[2,3]

Summary For the anterior and posterior fat pads to be useful diagnostic indicators on the lateral elbow, the elbow must be (1) flexed 90° and (2) in a true lateral position; (3) optimum exposure techniques, including soft tissue detail for visualization of fat pads, must be used.

Fig. 4.29 PA wrist—**scaphoid fat stripe (A).**

Fig. 4.30 PA oblique wrist—**scaphoid fat stripe (A).**

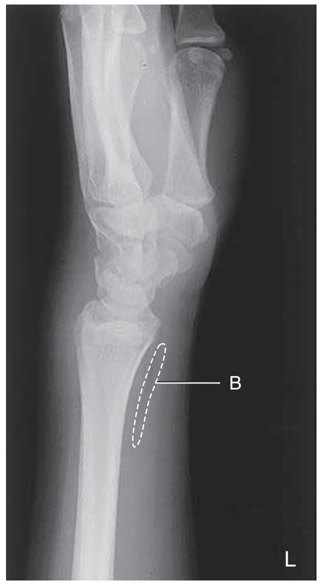


Fig. 4.31 Lateral wrist view—**pronator fat stripe (B).**

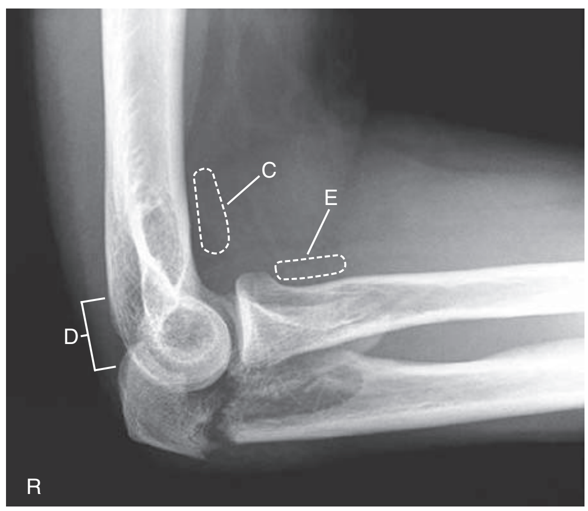


Fig. 4.32 Lateral elbow—fractured olecranon process (anterior and posterior fat pads), as follows: anterior fat pad (C); posterior fat pad (D), not visible; supinator fat stripe (E).

RADIOGRAPHIC POSITIONING

General Positioning Considerations

Radiographic examinations involving the upper limb on ambulatory patients generally are performed with the patient seated sideways at the end of the table, in a position that is neither strained nor uncomfortable. An extended tabletop may make this position more comfortable, especially if the patient is in a wheelchair. The patient's body should be moved away from the x-ray beam and the region of scatter radiation as much as possible. The height of the tabletop should be near shoulder height so that the arm can be fully supported (Fig. 4.33). The bucky tray should be moved to the opposite side of the radiographic table to reduce the amount of scatter radiation produced by the bucky device.

Lead Shielding

Shielding is important for examinations of the upper limb because of the proximity of the gonads to the divergent x-ray beam and scatter radiation, a risk for patients seated at the end of the table and for trauma patients taken on a stretcher. A lead, vinyl-covered shield should be draped over the patient's lap or gonadal area. Although the gonadal rule states that shielding should be provided to patients of reproductive age when the gonads lie within or close to the primary field, a good practice is to provide shielding for all patients.

Distance

A common minimum source to image receptor distance (SID) is 40 to 44 inches (100 to 110 cm). When radiographs are taken with the image receptor (IR) directly on the tabletop, to maintain a constant SID, the tube height must be increased compared with radiographs taken with the IR in the bucky tray. From the bucky tray to the tabletop, this difference is generally 3 to 4 inches (8 to 10 cm) for floating-type tabletops.

Special Patient Considerations

TRAUMA PATIENTS

Trauma patients can be radiographed on the table, or radiographs can be taken directly on the stretcher (Fig. 4.34). The patient should be moved to one side to provide the necessary space on the stretcher for the IR.

PEDIATRIC APPLICATIONS

Patient motion plays an important role in pediatric radiography. Immobilization is needed in many cases to assist children in maintaining the proper position. Sponges and tape are useful; however, sandbags should be used with caution because of their weight. Parents frequently are asked to assist with radiographic examination of their children. If parents are permitted in the radiography room during the exposure, proper shielding must be provided.

It is also important for the technologist to speak to the child in a soothing manner in language the child can readily understand to ensure maximal cooperation. (See Chapter 16 for more detailed explanations regarding upper limb radiography of young children.)

GERIATRIC APPLICATIONS

Providing **clear and complete instructions** is essential with elderly patients. Routine upper limb examinations may have to be altered to accommodate an older patient's physical condition. Geriatric patients may have greater difficulty in holding some of the strenuous positions required, so the technologist needs to ensure that adequate immobilization is used to prevent movement during the exposure. Radiographic exposure techniques may have to be reduced because of certain destructive pathologies commonly seen in elderly patients, such as osteoporosis.

Exposure Factors

The principal exposure factors for radiography of the upper limbs are as follows:

1. Lower to medium kVp (60 to 80—digital)
2. Short exposure time
3. Small focal spot
4. Adequate mAs for sufficient density (brightness)

Correctly exposed images of the upper limbs should reveal soft tissue margins for fat pad visualization and fine trabecular markings of all bones being radiographed.

Image Receptors

Grids are not generally used for the upper limb examinations unless the body part (e.g., the shoulder) measures greater than 10 cm.

Fig. 4.33 Ambulatory patient—lateral wrist (lead shield across lap covering gonads).

Fig. 4.34 Trauma patient—AP forearm.

TABLE 4.3 **CAST CONVERSION CHART**

TYPE OF CAST	INCREASE IN EXPOSURE
Small to medium plaster cast	Increase 5 to 7 kVp
Large plaster cast	Increase 8 to 10 kVp
Fiberglass cast	Increase 3 to 4 kVp

Increased Exposure with Casts

An upper limb with a cast requires an increase in exposure (Table 4.3). This increase depends on the thickness and type of cast, as outlined in the table. These increases apply to analog imaging. Digital imaging does not require exposure adjustments in most cases.

Collimation, General Positioning, and Markers

The collimation rule should be followed: Collimation borders should be visible on all four sides if the IR is large enough to allow this without cutting off essential anatomy.

A general rule regarding IR size is to use the smallest possible receptor size for the specific part that is being imaged. Four-sided collimation is generally possible even with an IR of minimum size for most, if not all, radiographic examinations of the upper limb.

A general positioning rule that is especially applicable to the upper limbs is **always to place the long axis of the part being imaged parallel to the long axis of the portion of the IR being exposed.**

Side markers within the collimation borders must be demonstrated on each image.

Correct Centering

Accurate centering and alignment of the body part with the IR and the central ray (CR) are important for examinations of the upper limb to avoid shape and size distortion and to demonstrate narrow joint spaces clearly. The following three positioning principles should be remembered for upper limb examinations:

1. Part should be parallel to plane of IR.
2. CR should be 90° or perpendicular to part and IR, unless a specific CR angle is indicated.
3. CR should be directed to correct centering point.

Digital Imaging Considerations

Specific guidelines should be followed when upper limb images are acquired through digital imaging technology (computed radiography or digital radiography).

1. **Collimation:** In addition to the benefit of reducing radiation dose to the patient, collimation that is closely restricted to the part being examined is key in ensuring that the image processed by the computer is of optimal quality. Close collimation also allows the computer to provide accurate information regarding the exposure indicator.
2. **Accurate centering:** Because of the way the image plate reader scans the exposed imaging plate, it is important in digital imaging, as in all radiographic imaging, that the body part and the CR be accurately centered to the IR.
3. **Grid use with digital systems (computed radiography/digital radiography):** As already mentioned, grids generally are not used for body parts measuring 10 cm or less. However, with direct digital radiography, grids may be used if the grid is an integral part of the IR mechanism. In such cases, because it may be impractical and difficult to remove the grid, it may be left in place even for smaller body parts, such as for upper and lower limb examinations.

NOTE: Keep x-ray tube centered to grid to avoid cutoff.

4. **Evaluation of exposure indicator:** After the image has been processed and is ready for viewing, the image is critiqued for exposure accuracy. It must be checked for an acceptable exposure indicator to verify that the exposure factors used were in the correct range to ensure an optimal quality image with the least possible radiation dose to the patient.

EXPOSURE FACTORS

Digital imaging systems are known for their wide exposure latitude; they are able to process an acceptable image from a broad range of exposure factors (kVp and mAs). It is still important, however, that the ALARA principle (as low as reasonably achievable) be followed regarding exposure to the patient; the lowest exposure factors that produce an optimal image should be used. This principle includes using the highest possible kVp and the lowest mAs consistent with desirable image quality as viewed on a radiologist-type interpretation monitor. Insufficient mAs results in a noisy (grainy) image on an interpretation monitor, even though it may appear satisfactory on a workstation monitor. Optimal kVp will provide the proper penetration to demonstrate the bony cortex and bony trabecular markings.

Alternative Modalities and Procedures

ARTHROGRAPHY

Arthrography is commonly used to image tendinous, ligamentous, and capsular pathology associated with diarthrodial joints, such as the wrist, elbow, shoulder, and ankle. This procedure requires the use of a radiographic contrast medium injected into the joint capsule under sterile conditions (see Chapter 19).

CT AND MRI

Computed tomography (CT) and MRI often are used on upper limbs to evaluate soft tissue and skeletal involvement of lesions and soft tissue injuries. Sectional CT images are also excellent for determination of displacement and alignment relationships with certain fractures that may be difficult to visualize with conventional radiographs.

NUCLEAR MEDICINE

Nuclear medicine bone scans are useful for demonstrating osteomyelitis, metastatic bone lesions, stress fractures, and cellulitis. Nuclear medicine scans demonstrate the pathologic process within 24 hours of onset. Nuclear medicine is more sensitive than radiography because it assesses the **physiologic aspect** instead of the anatomic aspect.

Clinical Indications

Clinical indications that technologists should be most familiar with in relation to the upper limb include the following (not an inclusive list).

Bone metastases refers to transfer of disease or cancerous lesions from one organ or part that may not be directly connected. All malignant tumors have the ability to metastasize, or transfer malignant cells from one body part to another, through the bloodstream or lymphatic vessels or by direct extension. Metastases are the most common of malignant bone tumors.

Bursitis *(ber-sy-tis)* is inflammation of the bursae or fluid-filled sacs that enclose the joints; the process generally involves the formation of calcification in associated tendons,[4] which causes pain and limited joint movement.

Carpal *(kar′-pul)* **tunnel syndrome** is a common painful disorder of the wrist and hand that results from compression of the median nerve as it passes through the center of the wrist; it is most commonly found in middle-aged women.

Fracture *(frak′-chur)* is a break in the structure of bone caused by a force (direct or indirect).[4] Numerous types of fractures have been identified; these are named by the extent of fracture, direction of fracture lines, alignment of bone fragments, and integrity of overlying tissue (see Chapter 15 for additional trauma and fracture terminology). Some common examples are as follows:

- **Barton fracture:** Fracture and dislocation of the **posterior lip of the distal radius** involving the wrist joint.
- **Bennett fracture:** Fracture of the **base of the first metacarpal bone,** extending into the carpometacarpal joint, complicated by subluxation with some posterior displacement.
- **Boxer fracture:** Transverse fracture that extends through the **metacarpal neck;** most commonly seen in the **fifth metacarpal.**
- **Colles fracture:** Transverse fracture of the **distal radius** in which the distal fragment is **displaced posteriorly;** an associated ulnar styloid fracture is seen in 50% to 60% of cases.
- **Smith fracture:** Reverse of Colles fracture, or transverse fracture of the **distal radius** with the distal fragment displaced **anteriorly.**

Joint effusion refers to **accumulated fluid** (synovial or hemorrhagic) in the joint cavity. It is a sign of an underlying condition, such as fracture, dislocation, soft tissue damage, or inflammation.

Osteoarthritis *(os″-te-o-ar-thry′-tis)*, also known as **degenerative joint disease (DJD),** is a noninflammatory joint disease characterized by gradual deterioration of the articular cartilage with hypertrophic (enlarged or overgrown) bone formation. This is the most common type of arthritis and is considered a normal part of the aging process.

Osteomyelitis *(os″-te-o-my″-e-ly′-tis)* is a local or generalized **infection of bone or bone marrow** that may be caused by bacteria introduced by trauma or surgery. However, it is more commonly the result of an infection from a contiguous source, such as a diabetic foot ulcer.

Osteopetrosis *(os″-te-o-pe-tro′-sis)* is a hereditary disease marked by **abnormally dense bone.** It commonly occurs as a result of fracture of affected bone and may lead to obliteration of the marrow space. This condition is also known as *marble bone.*

Osteoporosis *(os″-te-o-po-ro′-sis)* refers to **reduction in the quantity of bone** or **atrophy** of skeletal tissue. It occurs in postmenopausal women and elderly men, resulting in bone trabeculae that are scanty and thin. Most fractures sustained by women older than 50 years are secondary to osteoporosis.

Paget disease (osteitis deformans) is a common chronic skeletal disease; it is characterized by bone destruction followed by a reparative process of overproduction of very dense yet soft bones that tend to fracture easily. It is most common in men older than age 40. The cause is unknown, but evidence suggests involvement of a viral infection. Paget disease can occur in any bone but most commonly affects the pelvis, femur, tibia, skull, vertebrae, and clavicle.[4]

Rheumatoid *(ru′-ma-toyd)* **arthritis** is a chronic systemic disease with inflammatory changes throughout the connective tissues; the earliest change is soft tissue swelling that is most prevalent around the ulnar styloid of the wrist. Early bone erosions typically occur first at the second and third MCP joints or the third PIP joint. Rheumatoid arthritis is three times more common in women than in men.

Skier's thumb is a sprain or tear of the **ulnar collateral ligament of the thumb** near the MCP joint of the hyperextended thumb. The sprain or tear may result from an injury such as falling on an outstretched arm and hand, which causes the thumb to be bent back toward the arm. (The PA stress projection of bilateral thumbs [Folio method] best demonstrates this condition.)

Tumors (neoplasms, bone neoplasia) are most often benign (noncancerous) but may be malignant (cancerous). CT and MRI are helpful in determining the type and exact location and size of the tumor. Specific types of tumors are listed on p. 139.

4

- Malignant bone tumors
  - **Multiple myeloma** is the most common **primary cancerous bone tumor.** Multiple myeloma generally affects persons between ages 40 and 70 years. As the name implies, these tumors occur in various parts of the body, arising from bone marrow or marrow plasma cells. Therefore, these are not truly exclusively bone tumors. They are highly malignant and usually are fatal within a few years. The typical radiographic appearance includes multiple "punched-out" osteolytic (loss of calcium in bone) lesions scattered throughout the affected bones.[4]
  - **Osteogenic sarcoma (osteosarcoma)** is the second most common type of **primary cancerous bone tumor** and generally affects persons aged 10 to 20 years but can occur at any age. It may develop in older persons with Paget disease.
  - **Ewing sarcoma** is a common primary **malignant bone tumor** in children and young adults that arises from bone marrow. Symptoms are similar to symptoms of osteomyelitis with low-grade fever and pain. Stratified new bone formation results in an "onion peel" appearance on radiographs. The prognosis is poor by the time Ewing sarcoma is evident on radiographs.
  - **Chondrosarcoma** is a slow-growing **malignant tumor of the cartilage.** The appearance is similar to that of other malignant tumors, but dense calcifications are often seen within the cartilaginous mass.
- **Benign bone or cartilaginous tumors (chondromas)**
  - **Enchondroma** is a slow-growing **benign cartilaginous tumor** most often found in small bones of the hands and feet of adolescents and young adults. Generally, enchondromas are well-defined, radiolucent-appearing tumors with a thin cortex that often lead to pathologic fracture with only minimal trauma.
  - **Osteochondroma** (exostosis) is the most common type of **benign bone tumor,** usually occurring in persons aged 10 to 20 years. Osteochondromas arise from the outer cortex with the tumor growing parallel to the bone, pointing away from the adjacent joint. These are most common at the knee but also occur on the pelvis and scapula of children or young adults.

Table 4.4 provides a summary of clinical indications.

Routine and Special Projections

Routine and special projections for the hand, wrist, forearm, elbow, and humerus are demonstrated and described on the following pages.

TABLE 4.4 **SUMMARY OF CLINICAL INDICATIONS**

CONDITION OR DISEASE	MOST COMMON RADIOGRAPHIC EXAMINATION	POSSIBLE RADIOGRAPHIC APPEARANCE	EXPOSURE FACTOR ADJUSTMENT[a]
Conditions not Requiring an Exposure Factor Adjustment			
Select optimal exposure factors			
Bursitis	AP and lateral joint	Fluid-filled joint space with possible calcification	
Carpal tunnel syndrome	PA and lateral wrist; Gaynor-Hart method Sonography	Possible calcification in carpal sulcus Enlargement of wrist ligaments and median nerve compression	
Fractures	AP and lateral of long bones; AP, lateral, and oblique if joint involved	Disruption in bony cortex with soft tissue swelling	
Joint effusion	AP and lateral joint	Fluid-filled joint cavity	
Osteoarthritis (DJD)	AP and lateral affected area	Narrowing of joint space with periosteal growths on joint margins	None or decrease (−) in severe cases
Osteomyelitis	AP and lateral affected bone; nuclear medicine bone scan	Soft tissue swelling and loss of fat pad detail visibility	Visualize soft tissue structures
"Skier's thumb" (ulnar collateral ligament injury)	PA bilateral stress projection thumbs (Folio method)	Widening of inner MCP joint space of thumb and increase in degrees of angle of MCP line	
Tumors (neoplasms)—malignant and benign	AP and lateral affected area	Appearance dependent on type and stage of tumor	
Conditions Requiring an Increased Exposure Factor Adjustment			
Osteopetrosis (marble bone)	AP and lateral long bone	Chalky white or opaque appearance with lack of distinction between the bony cortex and trabeculae	
Paget disease	AP and lateral affected area	Mixed areas of sclerotic and cortical thickening along with radiolucent lesions; "cotton wool" appearance	May require increase (+) in advanced stages
Conditions Requiring a Decreased Exposure Factor Adjustment			
Osteoporosis	AP and lateral affected area	Best visibility in distal extremities and joints as decrease in bone density (brightness); long bones demonstrating thin cortex	
Rheumatoid arthritis (RA)	AP and lateral hand/wrist. Brewerton method can detect early signs of RA in hands	Closed joint spaces with subluxation of MCP joints	Decrease (−)

AP, Anteroposterior; *DJD,* degenerative joint disease; *MCP,* metacarpophalangeal; *PA,* posteroanterior.

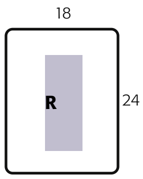
[a]Depends on stage or severity of disease or condition. Adjustments primarily apply to manual exposure factors.

PA PROJECTION—FINGERS

Clinical Indications
- Fractures and dislocations of the distal, middle, and proximal phalanges; distal metacarpal; and associated joints
- Pathologic processes, such as osteoporosis and osteoarthritis

Fingers
ROUTINE
- **PA**
- **PA oblique**
- **Lateral**

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



NOTE: A possible alternative routine involves a larger IR to include the entire hand for the PA projection of the finger for possible secondary trauma or pathology to other aspects of the hand and wrist. Then oblique and lateral projections of the affected finger only would be taken.

Fig. 4.35 PA—second digit.

Fig. 4.36 PA—fourth digit.

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow flexed about 90° and with hand and forearm resting on the table (Fig. 4.35).

Part Position
- Pronate hand with fingers extended
- Center and align long axis of affected finger with long axis of IR
- Separate adjoining fingers from affected finger (Fig. 4.36)

CR
- CR perpendicular to IR, directed to **PIP joint**

Recommended Collimation Collimate on four sides to area of affected finger and distal aspect of metacarpal

Fig. 4.37 PA—fourth digit.

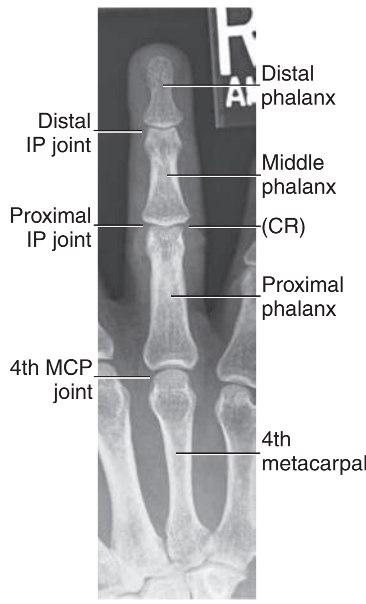


Fig. 4.38 PA—fourth digit.

Evaluation Criteria

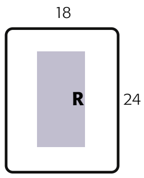
Anatomy Demonstrated: • Distal, middle, and proximal phalanges; distal metacarpal; and associated joints.
Position: • Long axis of finger should be aligned with and parallel to side border of IR. • **No rotation** of fingers is evidenced by symmetric appearance of both sides or concavities of the shafts of the phalanges and distal metacarpals. • The amount of tissue on each side of the phalanges should appear equal. • Fingers should be separated with no overlapping of soft tissues. • Interphalangeal joints should appear open, indicating that hand was fully pronated and the correct CR position was used (Figs. 4.37 and 4.38). • CR and midpoint of collimation field should be to the **PIP joint.**
Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

PA OBLIQUE PROJECTION—MEDIAL OR LATERAL ROTATION: FINGERS

Fingers
ROUTINE
- PA
- PA oblique
- Lateral

Clinical Indications
- Fractures and dislocations of the distal, middle, and proximal phalanges; distal metacarpal; and associated joints
- Pathologies such as osteoporosis and osteoarthritis

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65
- Accessories—45° foam wedge block or step wedge



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow flexed about 90° with hand and wrist resting on IR and fingers extended.

Part Position
- With fingers extended against 45° foam wedge block, place hand in a 45° lateral oblique (thumb side up) (Fig. 4.39).
- Position hand on image receptor so that the long axis of the finger is aligned with the long axis of the IR.
- Separate fingers and carefully place finger that is being examined against block, so it is supported in a 45° oblique and **parallel to IR.**

CR
- CR perpendicular to IR, directed to **PIP joint**

Recommended Collimation Collimate on four sides to affected finger and distal aspect of metacarpal.

Optional Medial Oblique Second digit also may be taken in a 45° medial oblique (thumb side down) with thumb and other fingers flexed to prevent superimposition (Fig. 4.40). This position places the part closer to the IR for improved definition but may be more painful for the patient. Lateral rotation of hand is recommended to demonstrate the third, fourth, and fifth digits (Figs. 4.41 and 4.42)


Fig. 4.39 Second digit (lateral rotation).


Fig. 4.40 Second digit (medial rotation).

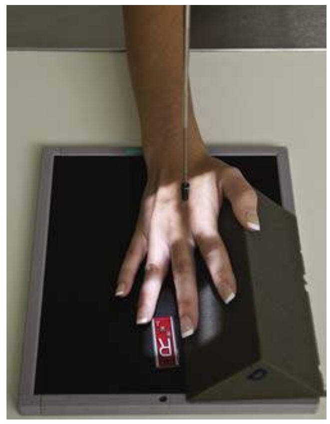


Fig. 4.41 Third digit (lateral rotation).

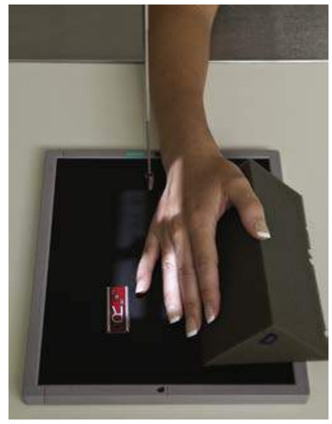


Fig. 4.42 Fifth digit (lateral rotation).


Fig. 4.43 Fourth digit (lateral rotation).

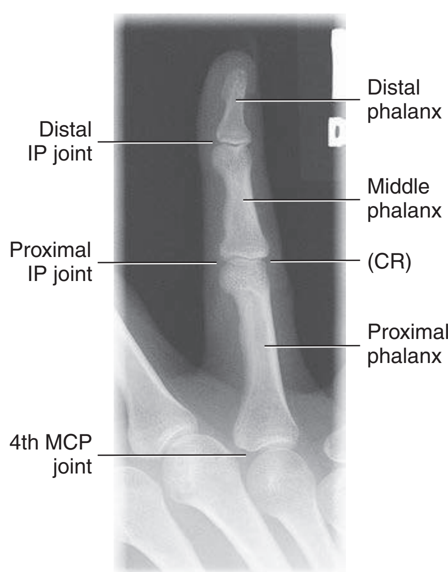


Fig. 4.44 Fourth digit.

Evaluation Criteria

Anatomy Demonstrated: • Oblique view of distal, middle, and proximal phalanges; distal metacarpal; and associated joints.

Position: • Long axis of finger should be aligned with side border of IR. • View of finger being examined should be 45° oblique. • No superimposition of adjacent fingers should occur. • IP and MCP joint spaces should be open, indicating correct CR location and that the phalanges are parallel to IR. • CR and center of collimation field should be to the **PIP joint** (Figs. 4.43 and 4.44).

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

LATEROMEDIAL OR MEDIOLATERAL PROJECTIONS—FINGERS

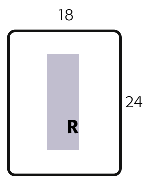
Clinical Indications

- Fractures and dislocations of the distal, middle, and proximal phalanges; distal metacarpal; and associated joints
- Pathologic processes, such as osteoporosis and osteoarthritis

Fingers
ROUTINE
- PA
- PA oblique
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65
- Accessories—sponge support block



Shielding Shield radiosensitive tissues outside region of interest

Patient Position Seat patient at end of table, with elbow flexed about 90° with hand and wrist resting on IR and fingers extended.

Part Position

- Place hand in lateral position (thumb side up) with finger to be examined fully extended and centered to portion of IR being exposed (see NOTE for second digit lateral).
- Align and center finger to long axis of IR and to CR.
- Use sponge block or other radiolucent device to support finger and prevent motion. Flex unaffected fingers (Fig. 4.45).
- Ensure that long axis of finger is **parallel to IR** (Figs. 4.46 to 4.48).

CR

- CR perpendicular to IR, directed to **PIP joint**

Recommended Collimation Collimate on four sides to affected finger and distal aspect of metacarpal

NOTE: For second digit, a mediolateral is advised (Fig. 4.45) if the patient can assume this position. Place the second digit in contact with IR. (Definition is improved with less object–image receptor distance [OID].)

Evaluation Criteria

Anatomy Demonstrated: • Lateral views of distal, middle, and proximal phalanges; distal metacarpal; and associated joints are visible (Figs. 4.49 and 4.50).

Position: • Long axis of finger should be aligned with the side border of IR. • Finger should be in **true lateral position,** as indicated by the concave appearance of the anterior surface of the shaft of the phalanges. • Interphalangeal and metacarpophalangeal joint spaces should be open, indicating correct CR location and that the phalanges are parallel to the IR. • CR and center of collimation field should be to the **PIP joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

Fig. 4.45 Second digit (mediolateral).

Fig. 4.46 Third digit (lateromedial).

Fig. 4.47 Fourth digit (lateromedial).

Fig. 4.48 Fourth digit lateromedial.

Fig. 4.49 Fourth digit.

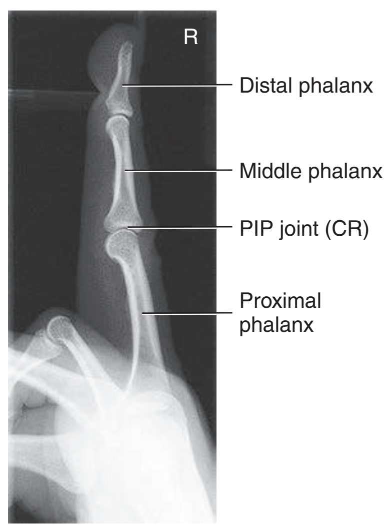


Fig. 4.50 Fourth digit.

4

AP PROJECTION—THUMB

Clinical Indications

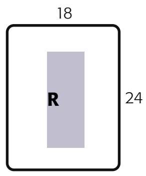
- Fractures and dislocations of the distal and proximal phalanges, distal metacarpal, and associated joints
- Pathologic processes, such as osteoporosis and osteoarthritis

See special AP modified Robert projection for Bennett fracture at base of first metacarpal.

Thumb
ROUTINE
• AP
• PA oblique
• Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position—AP Seat patient facing table, arms extended in front, with hand rotated internally to supinate thumb for AP projection (Fig. 4.51).

Part Position—AP

First, demonstrate this awkward position on yourself, so the patient can see how it is done and better understand what is expected.

- Internally rotate hand with fingers extended until posterior surface of thumb is in contact with IR. Immobilize other fingers with tape to isolate thumb if necessary (see Fig. 4.51).
- Align thumb with long axis of the IR.
- Center **first MCP joint** to CR and to center of IR.

Exception—PA (Only if Patient Cannot Position for Previous AP)

- Place hand in near-lateral position and rest thumb on sponge support block that is high enough so the thumb is not rotated but is in position for a **true PA projection** (Fig. 4.52)

NOTE: As a rule, the PA is *not* advisable because it results in loss of definition caused by increased OID.

CR

- CR perpendicular to IR, to **first MCP joint**

Recommended Collimation

Collimate on four sides to area of thumb, remembering that **thumb includes entire first metacarpal and trapezium**

Fig. 4.51 AP thumb—CR to first MCP joint.

Fig. 4.52 PA (exception).

Fig. 4.53 AP thumb.

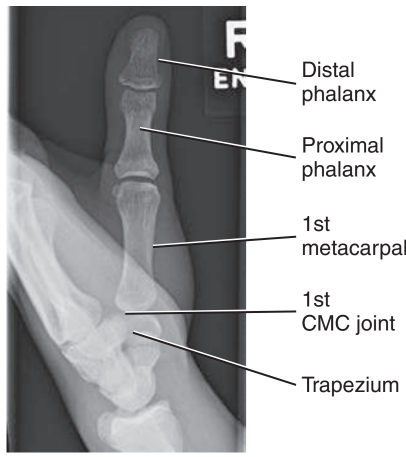


Fig. 4.54 AP thumb.

Evaluation Criteria

Anatomy Demonstrated: • Distal and proximal phalanges, first metacarpal, trapezium, and associated joints are visible. • Interphalangeal and metacarpophalangeal joints should appear open (Figs. 4.53 and 4.54).

Position: • Long axis of thumb should be aligned with side border of IR. • **No rotation,** as evidenced by the concave sides of the phalanges and by equal amounts of soft tissue appearing on each side of the phalanges, should be present. Interphalangeal joint should appear open, indicating that thumb was fully extended and correct CR location was used. • CR and center of collimation field should be at the **first MCP joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

PA OBLIQUE PROJECTION—MEDIAL ROTATION: THUMB

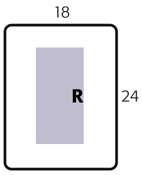
Clinical Indications

- Fractures and dislocations of the distal and proximal phalanges, distal metacarpal, and associated joints
- Pathologic processes, such as osteoporosis and osteoarthritis

Thumb
ROUTINE
- AP
- PA oblique
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand resting on IR.

Part Position

- Abduct thumb slightly with palmar surface of hand in contact with IR (this action naturally places thumb in a 45° oblique position).
- Align long axis of thumb with long axis of IR.
- Center **first MCP joint** to CR and to center of IR (Fig. 4.55).

CR

- CR perpendicular to IR, directed to **first MCP joint**

Recommended Collimation Collimate on four sides to thumb, ensuring that all of first metacarpal and trapezium is included.

Fig. 4.55 PA oblique thumb—CR to first MCP joint.

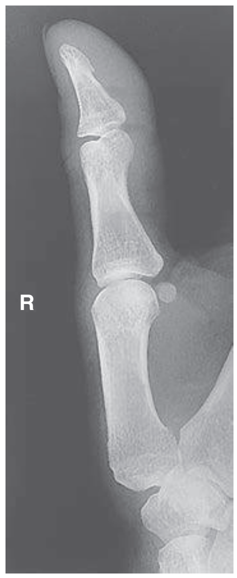


Fig. 4.56 PA oblique thumb.

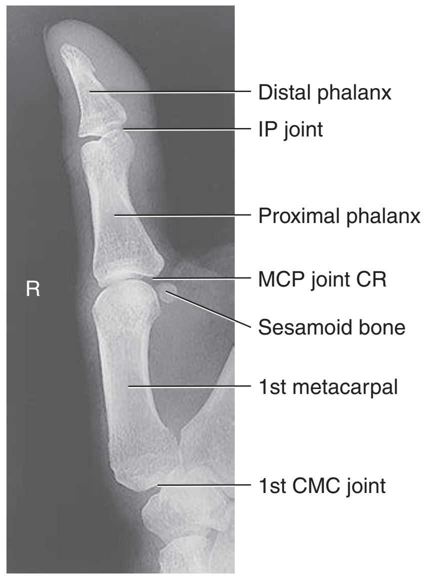


Fig. 4.57 PA oblique thumb.

Evaluation Criteria

Anatomy Demonstrated: • Distal and proximal phalanges, first metacarpal, trapezium, and associated joints are visualized in a 45° oblique position (Figs. 4.56 and 4.57).
Position: • Long axis of thumb should be aligned with side border of IR. • Interphalangeal and metacarpophalangeal joints should appear open if the phalanges are parallel to the IR and if the CR location is correct. • CR and center of collimation field should be at **first MCP joint.**
Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

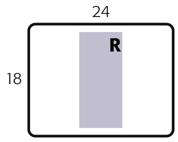
4

LATERAL POSITION—THUMB

Thumb
ROUTINE
- AP
- PA oblique
- Lateral

Clinical Indications
- Fractures and dislocations of the distal and proximal phalanges, distal metacarpal, and associated joints
- Pathologic processes, such as osteoporosis and osteoarthritis

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow flexed about 90° with hand resting on IR, palm down.

Part Position
- Start with hand pronated and thumb abducted, with fingers and hand slightly arched; then rotate hand slightly medial until thumb is in **true lateral position** (You may need to provide a sponge or other support under lateral portion of hand)
- Align long axis of thumb with long axis of the IR.
- Center **first MCP joint** to CR and to center of IR.
- Entire lateral aspect of thumb should be in direct contact with IR (Fig. 4.58)

CR
- CR perpendicular to IR, directed to **first MCP joint**

Recommended Collimation Collimate on four sides to thumb area (remember that first metacarpal and **trapezium** must be within the field of view).

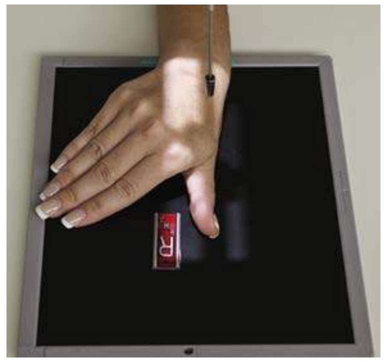


Fig. 4.58 Part position—lateral thumb; CR to first MCP joint.

Fig. 4.59 Lateral thumb.

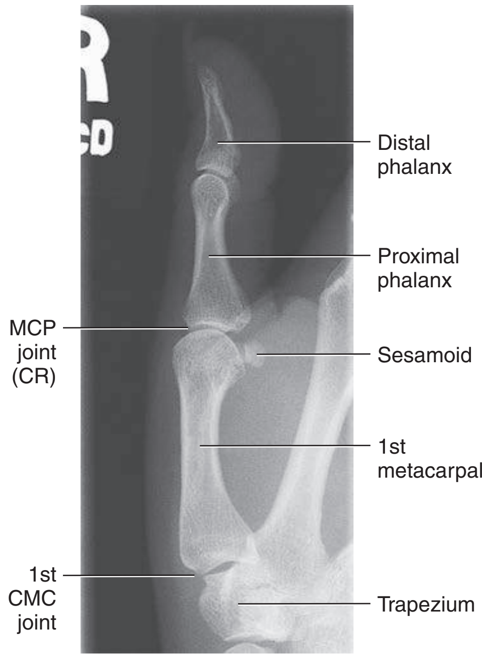


Fig. 4.60 Lateral thumb.

Evaluation Criteria
Anatomy Demonstrated: • Distal and proximal phalanges, first metacarpal, trapezium (superimposed), and associated joints are visualized in the lateral position (Figs. 4.59 and 4.60).
Position: • Long axis of thumb should be aligned with side border of IR. • Thumb should be in a true lateral position, evidenced by the concave-shaped anterior surface of the proximal phalanx and first metacarpal and relatively straight posterior surfaces. • Interphalangeal and metacarpophalangeal joints should appear open if the phalanges are parallel to the IR and if the CR location is correct. • CR and center of collimation field should be at the **first MCP joint.**
Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

AP AXIAL PROJECTION (MODIFIED ROBERT METHOD)[5]—THUMB

Thumb
SPECIAL
- AP axial, modified Robert method

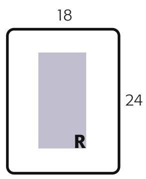
Clinical Indications

- Base of first metacarpal is demonstrated for ruling out **Bennett fracture.**

This special projection demonstrates fractures, dislocations, or pathology of the base of the first metacarpal and trapezium

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient parallel to end of table, with hand and arm fully extended.

Part Position

- Rotate arm internally until posterior aspect of thumb rests on IR.
- Place thumb in center of IR, parallel to side border of IR.
- Extend fingers.

CR

- CR directed **15° proximally** (toward wrist), entering at the **first CMC joint**
- Lewis modification—CR angle 10° to 15° proximal to MCP joint (see NOTE**)**

Recommended Collimation Collimate on four sides to area of thumb and first CMC joint.

NOTE: This projection was first described by M. Robert in 1936 to demonstrate the first CMC joint with the use of a **perpendicular** CR. The projection was later modified to include **15°** proximal CR angle to the first CMC joint.[6] The **Lewis modification** centers the CR to the first MCP joint with a **10° to 15°** proximal angle[7] (Fig. 4.61).

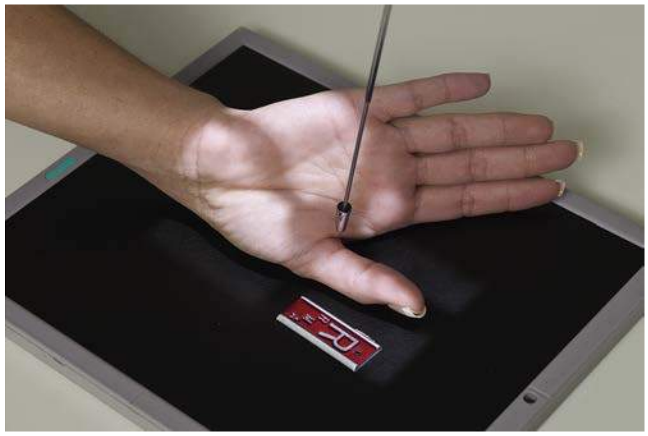


Fig. 4.61 AP axial projection—Lewis modification, CR 10° to15° to MCP joint.

Fig. 4.62 AP axial projection—Lewis modification.

Fig. 4.63 AP axial projection—modified Robert method.

Evaluation Criteria

Anatomy Demonstrated: • An AP projection of the thumb and first CMC joint are visible without superimposition. • Base of first metacarpal and trapezium should be well visualized (Figs. 4.62 and 4.63).

Position: • Long axis of the thumb should be aligned with side border of IR. • **No rotation,** as evidenced by the symmetric appearance of both concave sides of the phalanges and by the equal amounts of soft tissue that appear on each side of the phalanges. • First CMC and MCP joints should appear open. • CR and center of collimation field should be at **first CMC joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

PA STRESS THUMB PROJECTION

FOLIO METHOD[8]

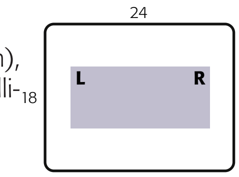
Thumb
SPECIAL
• AP axial, modified Robert method
• PA stress (Folio method)

Clinical Indications

- Sprain or tearing of ulnar collateral ligament of thumb at MCP joint as a result of acute hyperextension of thumb; also referred to as a "skier's thumb" injury

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with both hands extended and pronated on IR.

Part Position

- Position both hands side by side to center of IR, rotated laterally into ±45° oblique position, resulting in PA projection of both thumbs.
- Place supports as needed under both wrist and proximal thumb regions to prevent motion. Ensure that hands are rotated enough to place thumbs parallel to IR for **PA projection** of both thumbs.
- Place round spacer, such as a roll of medical tape, between proximal thumb regions; wrap rubber bands around distal thumbs (Fig. 4.64).
- Immediately before exposure, ask patient to pull thumbs apart firmly and hold.

NOTE: Explain procedure carefully to patient and observe patient while applying tension on rubber band without motion before initiating exposure. Work quickly because this can be painful for patient.

CR

- CR perpendicular to IR directed to midway between MCP joints

Recommended Collimation Collimate on four sides to include second metacarpals and entire thumbs, from CMC joints proximally to distal phalanges distally.

Evaluation Criteria

Anatomy Demonstrated: • Entire thumbs from first metacarpals to distal phalanges (Fig. 4.65). • Demonstrates metacarpophalangeal angles and joint spaces at MCP joints (Fig. 4.66).

Position: • **No rotation** of thumbs as evidenced by symmetric appearance of concavities of shafts of first metacarpals and phalanges. • Distal phalanges should appear to be pulled together, indicating that tension was applied. • MCP and IP joints should appear open, indicating that thumbs were parallel to IR and perpendicular to CR. • CR and center of collimation field should be **midway between the two MCP joints.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrates soft tissue margins and clear, sharp bony edges and trabecular markings.

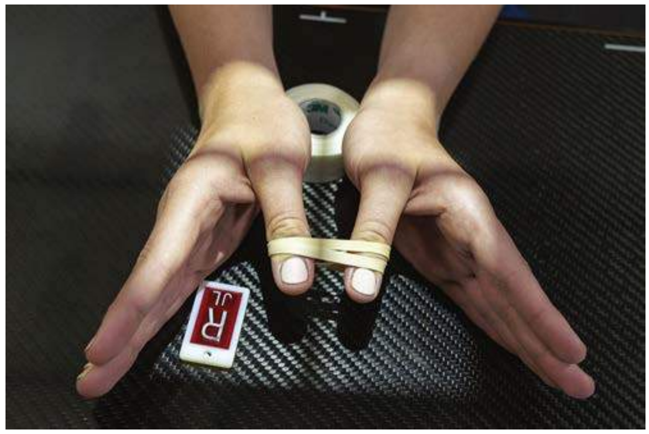


Fig. 4.64 PA stress projection of bilateral thumbs; CR perpendicular to midway between MCP joints, firm tension applied.

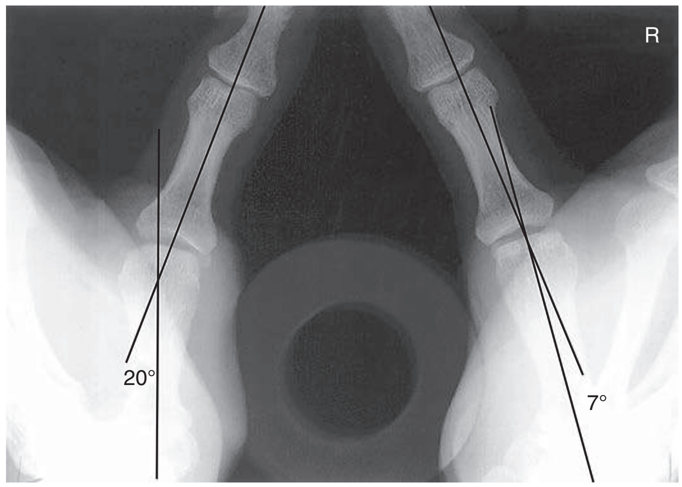


Fig. 4.65 PA stress projection of bilateral thumbs with tension applied. 20° MCP angle on left indicates sprain or torn ulnar collateral ligament. (From Frank ED, Long BW, Smith BJ. *Merrill's atlas of radiographic positions and radiologic procedures,* ed 11, St Louis, 2007, Mosby.)

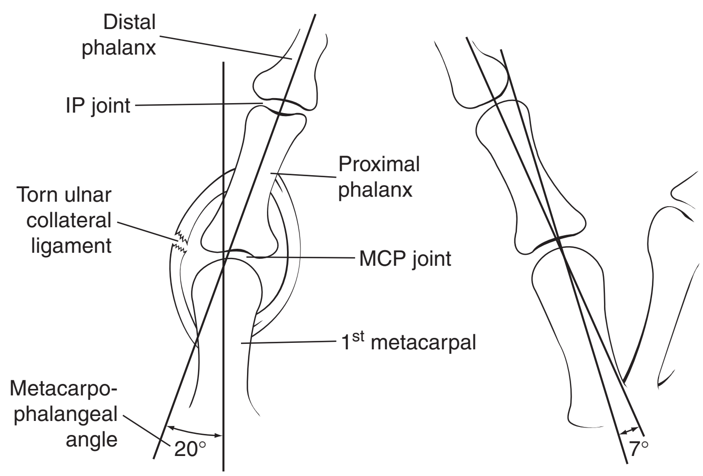


Fig. 4.66 PA stress projection of bilateral thumbs with tension applied (demonstrates torn ulnar collateral ligament on left).

PA PROJECTION—HAND

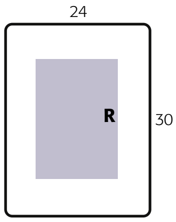
Clinical Indications

- Fractures, dislocations, or foreign bodies of the phalanges, metacarpals, and all joints of the hand
- Pathologic processes such as osteoporosis and osteoarthritis

Hand
ROUTINE
- **PA**
- PA oblique
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended.

Part Position

- Pronate hand with palmar surface in contact with IR; spread fingers slightly (Fig. 4.67)
- Align long axis of hand and forearm with long axis of IR.
- Center hand and wrist to IR

CR

- CR perpendicular to IR, directed to **third MCP joint**

Recommended Collimation Collimate on four sides to outer margins of hand and wrist.

NOTE: If examinations of both hands or wrists are requested, generally the body parts should be positioned and exposed separately for correct CR placement.

Evaluation Criteria

Anatomy Demonstrated: • PA projection of entire hand and wrist and about 1 inch (2.5 cm) of distal forearm are visible. • PA projection of hand demonstrates oblique view of the thumb.

Position: • Long axis of hand and wrist aligned with long axis of IR. • **No rotation** of hand, as evidenced by symmetric appearance of both sides or concavities of shafts of metacarpals and phalanges of digits 2 through 5 and the appearance of equal amounts of soft tissue on each side of phalanges 2 through 5. • Digits should be separated slightly with soft tissues not overlapping. • MCP and IP joints should appear open, indicating correct CR location and that hand was fully pronated (Figs. 4.68 and 4.69). • CR and center of collimation field should be to **third MCP joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

Fig. 4.67 PA hand, CR to third MCP joint.

Fig. 4.68 PA hand.

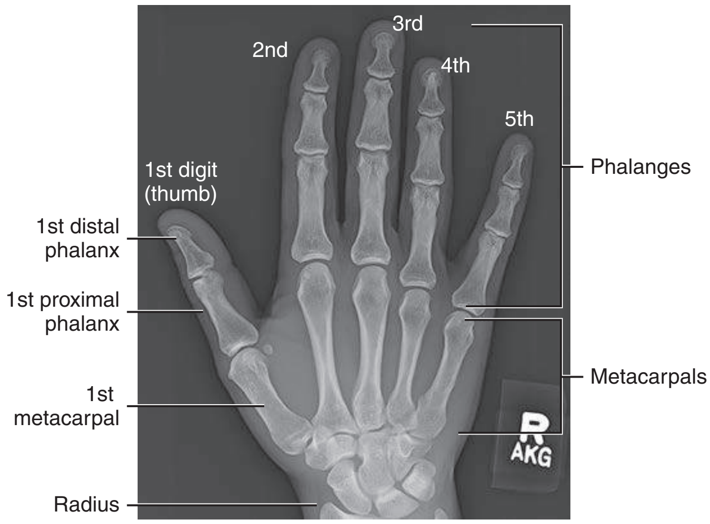


Fig. 4.69 PA of right hand.

PA OBLIQUE PROJECTION—HAND

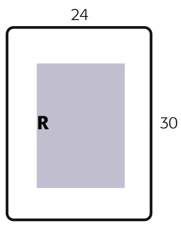
Clinical Indications

- Fractures and dislocations of the phalanges, metacarpals, and all joints of the hand
- Pathologic processes, such as osteoporosis and osteoarthritis

Hand
ROUTINE
- PA
- **PA oblique**
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended.

Part Position

- Pronate hand on IR; center and align long axis of hand with long axis of IR.
- Rotate entire hand and wrist laterally 45° and support with radiolucent wedge or step block, as shown, so that all digits are separated and **parallel to IR** (see Exception).

CR

- CR perpendicular to IR, directed to **third MCP joint**

Recommended Collimation Collimate on four sides to hand and wrist.

Exception For a routine oblique hand, use a support block to place digits parallel to IR (Fig. 4.70). This block prevents foreshortening of phalanges and obscuring of interphalangeal joints. If the **metacarpals only** are of interest, the image can be taken with thumb and fingertips touching IR (Figs. 4.71 and 4.73).

Fig. 4.70 Routine oblique hand (digits parallel).

Fig. 4.71 *Exception:* Oblique hand for metacarpals (digits not parallel)—not recommended for digits.

Fig. 4.72 PA oblique hand (digits parallel).

Fig. 4.73 PA oblique hand (digits not parallel)—joint spaces not open.

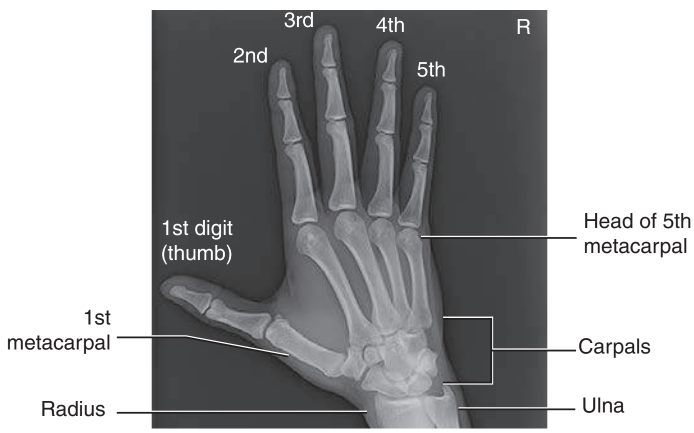


Fig. 4.74 PA oblique hand (digits parallel).

Evaluation Criteria

Anatomy Demonstrated: • Oblique projection of the entire hand and wrist and about 1 inch (2.5 cm) of distal forearm are visible.

Position: • Long axis of hand and wrist should be aligned with IR. • 45° oblique is evidenced by: midshafts of metacarpals should not overlap; some overlap of distal heads of third, fourth, and fifth metacarpals but no overlap of distal second and third metacarpals should occur; excessive overlap of metacarpals indicates over-rotation, and too much separation indicates under-rotation. • MCP and IP joints are open without foreshortening of midphalanges or distal phalanges, indicating that fingers are parallel to IR (Figs. 4.72 and 4.74). • CR and center of collimation field should be at **third MCP joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings.

4

"FAN" LATERAL—LATEROMEDIAL PROJECTION: HAND

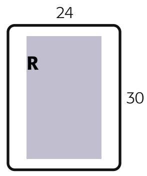
Clinical Indications

- Fractures and dislocations of the phalanges, anterior/posterior displaced fractures, and dislocations of the metacarpals
- Pathologic processes, such as osteoporosis and osteoarthritis especially in the phalanges

Hand
ROUTINE
• PA
• PA oblique
• **Lateral**

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65
- Accessories—45° foam step support



Compensation Filter A filter may be used to ensure optimum exposure of phalanges and metacarpals because of differences in part thickness.

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended.

Part Position

- Align long axis of hand with long axis of IR.
- Rotate hand and wrist into lateral position with thumb side up.
- Spread fingers and thumb into a "fan" position, and support each digit on radiolucent block as shown. Ensure that all digits, including the thumb, are separated and **parallel to IR** and that the metacarpals are *not* rotated but remain in a true lateral position (Fig. 4.75)

CR

- CR perpendicular to IR, directed to **second MCP joint**

Recommended Collimation Collimate on four sides to outer margins of hand and wrist.

NOTE: The "fan" lateral position is the preferred lateral for the hand if phalanges are the area of interest. (See next page for alternative projections.)

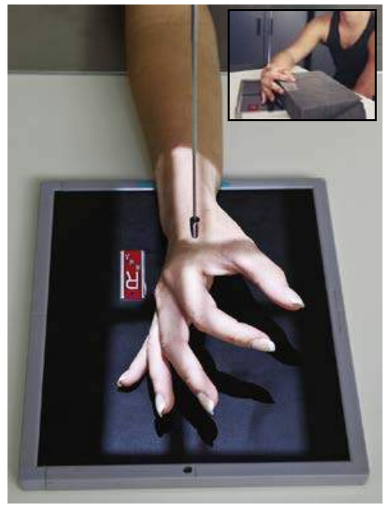


Fig. 4.75 Patient position—fan lateral hand (digits kept separated and parallel to IR); CR to second MCP joint.

Fig. 4.76 Fan lateral projection.

4

Evaluation Criteria

Anatomy Demonstrated: • Entire hand and wrist and about 1 inch (2.5 cm) of distal forearm are visible (Figs. 4.76 and 4.77).

Position: • Long axis of hand and wrist should be aligned with long axis of IR. • Fingers should appear equally separated, with phalanges in the lateral position and joint spaces open, indicating that fingers were parallel to IR. • Thumb should appear in slightly oblique position completely free of superimposition, with joint spaces open. • Hand and wrist should be in a true lateral position, as evidenced by: distal radius and ulna are superimposed; metacarpals are superimposed. • CR and center of collimation field should be at **second MCP joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings. • Outlines of individual metacarpals demonstrated are superimposed. • Midphalanges and distal phalanges of thumb and fingers should appear sharp but may be slightly overexposed.

Fig. 4.77 Fan lateral projection of right hand.

LATERAL IN EXTENSION AND FLEXION—LATEROMEDIAL PROJECTIONS: HAND

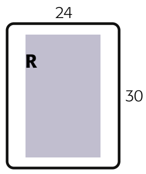
ALTERNATIVES TO FAN LATERAL

Clinical Indications

Hand
ALTERNATE
• Extension lateral
• Flexion lateral

- The lateral in either extension or flexion is an alternative to the fan lateral for localization of foreign bodies of the hand and fingers; it also demonstrates anterior or posterior displaced fractures of the metacarpals.
  The lateral in a natural flexed position may be less painful for the patient.

Technical Factors



- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended.

Part Position
Rotate hand and wrist, with thumb side up, into **true lateral position,** with second to fifth MCP joints centered to IR and CR.
- **Lateral in extension:** Extend fingers and thumb, and support against a radiolucent support block. Ensure that all fingers and metacarpals are superimposed directly for true lateral position (Fig. 4.78)
- **Lateral in flexion:** Flex fingers into a natural flexed position, with thumb lightly touching the first finger; maintain true lateral position (Fig. 4.79)

CR
- CR perpendicular to IR, directed to the **second to fifth MCP joints**

Recommended Collimation Collimate to outer margins of hand and wrist.

Evaluation Criteria
Anatomy Demonstrated: • Entire hand and wrist and about 1 inch (2.5 cm) of distal forearm are visible. • Thumb should appear in slightly oblique position and free of superimposition with joint spaces open.
Position: • Long axis of the hand and wrist is aligned with long axis of the IR. • Hand and wrist should be in **true lateral position,** as evidenced by: distal radius and ulna are superimposed; metacarpals and phalanges are superimposed. • **Lateral in extension**: The phalanges and metacarpals should be superimposed and extended (Fig. 4.80). • **Lateral in flexion**: The phalanges and metacarpals should be superimposed with hand in natural flexed position (Figs. 4.81 and 4.82). • CR and center of collimation field should be at **second to fifth MCP joints.**
Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate soft tissue margins and clear, sharp bony trabecular markings. • Margins of individual metacarpals and phalanges are visible but mostly superimposed.

Fig. 4.78 Lateral in extension.

Fig. 4.79 Lateral in flexion.

Fig. 4.80 Lateral in extension.

Fig. 4.81 Lateral in flexion.

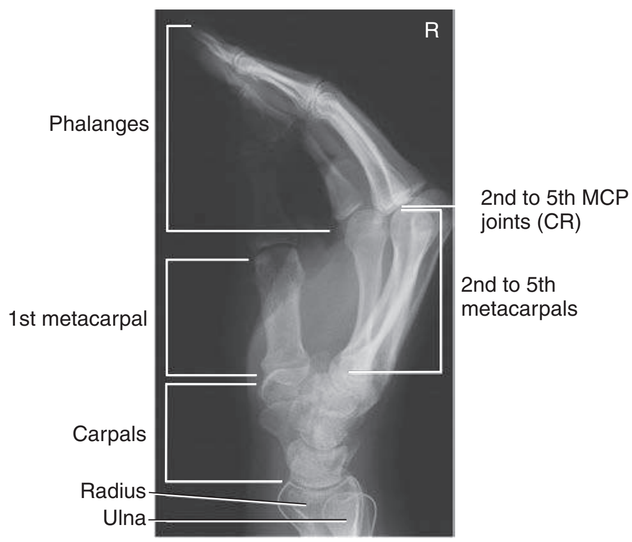


Fig. 4.82 Lateral in flexion.

4

AP AXIAL PROJECTION—HAND

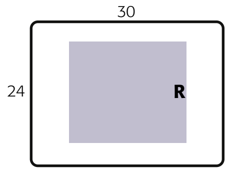
BREWERTON METHOD[9,10]

Clinical Indications

Hand
SPECIAL
- AP Axial

- Performed commonly to evaluate for early evidence of rheumatoid arthritis at the second through fifth metacarpophalangeal (MCP) joints. Evident by slight erosion of the head of the metacarpal.
- May demonstrate fractures of the base of the fourth and fifth metacarpal.

Technical Factors



- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm) portrait or 14 × 17 inches (35 × 43 cm) for bilateral study, landscape; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Stand patient at end of table with hand supinated and flexed.

Part Position

- Supinate hand and place at the center of the IR.
- From this position, keeping fingers in contact with the IR, flex the hand to create a 65° angle between the dorsum of hand and IR (Fig. 4.83).
- Extend fingers and ensure they are relaxed, slightly separated and parallel to IR.
- Abduct thumb to avoid superimposition.

CR

- Angle CR 15° proximally, toward ulna, directed to the **third MCP joint.**

Recommended Collimation Collimate on four sides to outer margins of hand and wrist.

4

Evaluation Criteria

Anatomy Demonstrated: • Entire hand from the carpal area to the tips of digits are visible (Figs. 4.84 and 4.85).
Position: • Second through fifth MCP joints open and visible with no superimposition of the palmer soft tissue; thumb should be free of superimposition with second through fifth digits. Midshafts of second through fifth metacarpals and phalanges should not overlap • CR and center of collimation field should be **at the third MCP joint.**
Exposure: • Optimal density (brightness) and contrast with **no motion** are demonstrated by clear, sharp bony trabecular markings and joint space margins of MCP joints.

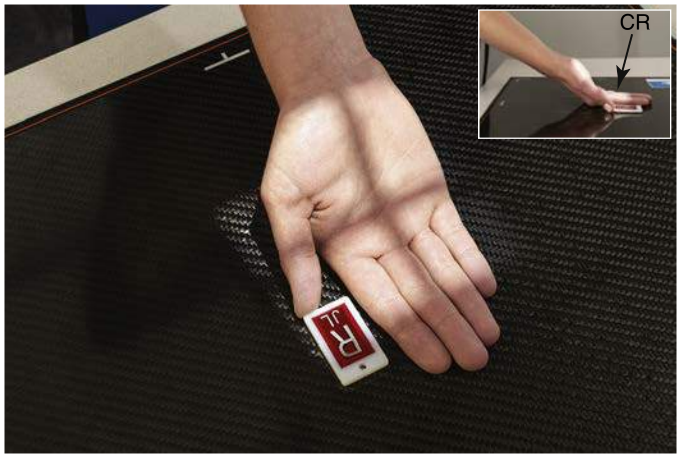


Fig. 4.83 AP axial—Brewerton method.

Fig. 4.84 AP axial. (From Wilson DJ et al. *Musculoskeletal imaging*, ed 2, Philadelphia, 2015, Elsevier.)

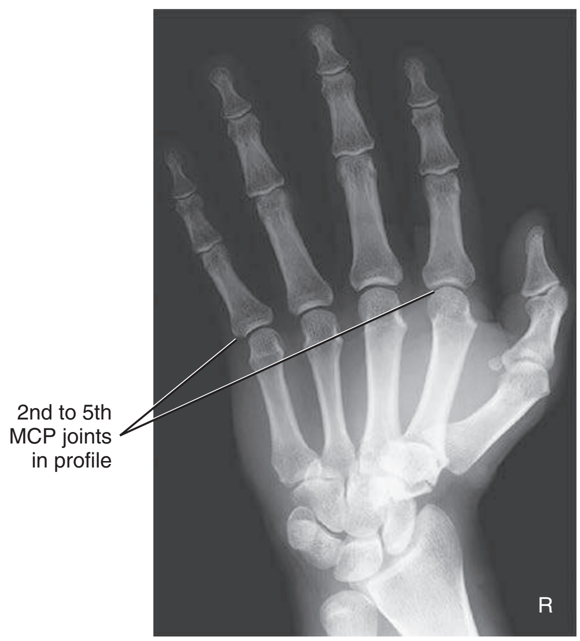


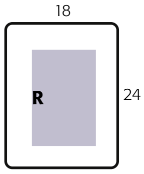
Fig. 4.85 AP axial. (Modified from Wilson DJ et al. *Musculoskeletal imaging*, ed 2, Philadelphia, 2015, Elsevier.)

PA (AP) PROJECTION—WRIST

Wrist
ROUTINE
• PA
• PA oblique
• Lateral

Clinical Indications

- Fractures of distal radius or ulna, isolated fractures of radial or ulnar styloid processes, and fractures of individual carpal bones
- Pathologic processes, such as osteomyelitis and arthritis



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended. Drop shoulder so that shoulder, elbow, and wrist are on same horizontal plane.

Part Position

- Align and center long axis of hand and wrist to IR, with carpal area centered to CR.
- With hand pronated, arch hand slightly to **place wrist and carpal area in close contact with IR** (Fig. 4.86).

CR

- CR perpendicular to IR, directed to **midcarpal area**

Recommended Collimation Collimate to wrist on all four sides; include distal radius and ulna and midmetacarpal area.

Alternative AP An AP wrist may be taken, with hand slightly arched to place **wrist and carpals in close contact with IR,** to demonstrate intercarpal spaces and wrist joint better and to place the intercarpal spaces more parallel to the divergent rays (Fig. 4.87). This wrist projection is good for visualizing the carpals if the patient can assume this position easily.

Evaluation Criteria

Anatomy Demonstrated: • Midmetacarpals and proximal metacarpals; carpals; distal radius, ulna, and associated joints; and pertinent soft tissues of the wrist joint, such as fat pads and fat stripes, are visible. • All the intercarpal spaces do not appear open because of irregular shapes that result in overlapping (Figs. 4.88 and 4.89).

Position: • Long axis of the hand, wrist, and forearm is aligned with IR. • True PA is evidenced by: equal concavity shapes are on each side of the shafts of the proximal metacarpals; near-equal distances exist among the proximal metacarpals; separation of the distal radius and ulna is present except for possible minimal superimposition at the distal radioulnar joint. • CR and center of collimation field should be to the **midcarpal area.**

Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize soft tissue, such as pertinent fat pads, and sharp, bony margins of the carpals and clear trabecular markings.

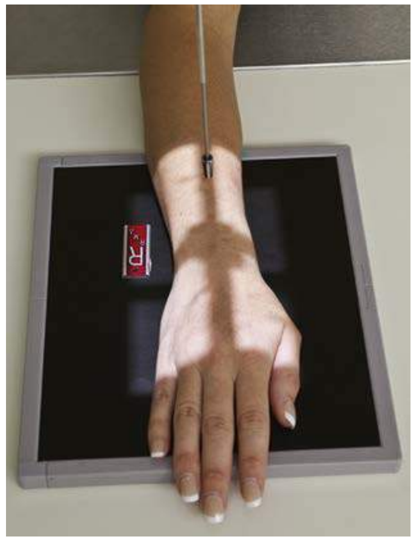


Fig. 4.86 PA wrist.

Fig. 4.87 Alternative AP wrist.

Fig. 4.88 PA wrist.

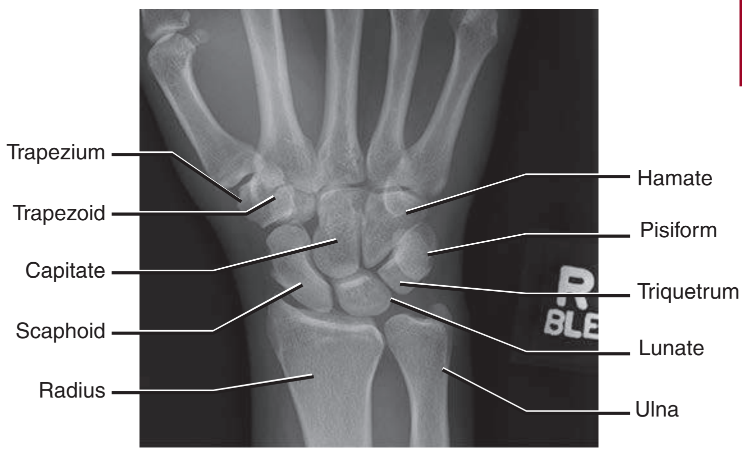


Fig. 4.89 PA of right wrist.

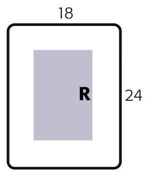
PA OBLIQUE PROJECTION—LATERAL ROTATION: WRIST

Wrist
ROUTINE
- PA
- PA oblique
- Lateral

Clinical Indications

- Fractures of distal radius or ulna, isolated fractures of radial or ulnar styloid processes, and fractures of individual carpal bones
- Pathologic processes, such as osteomyelitis and arthritis

Technical Factors



- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—60 to 70

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended. Drop shoulder so that shoulder, elbow, and wrist are on same horizontal plane.

Part Position

- Align and center hand and wrist to IR.
- From pronated position, rotate wrist and hand laterally 45°.
- For stability, place a 45° support under thumb side of hand to support hand and wrist in a 45° oblique position (Fig. 4.90) or partially flex fingers to arch hand so that fingertips rest lightly on IR without support (Fig. 4.91).

CR

- CR perpendicular to IR, directed to **midcarpal area**

Recommended Collimation Collimate to wrist on four sides; include distal radius and ulna and midmetacarpal area.

Fig. 4.90 PA oblique wrist (with 45° support).

Fig. 4.91 PA oblique wrist without support.

Fig. 4.92 PA oblique wrist.

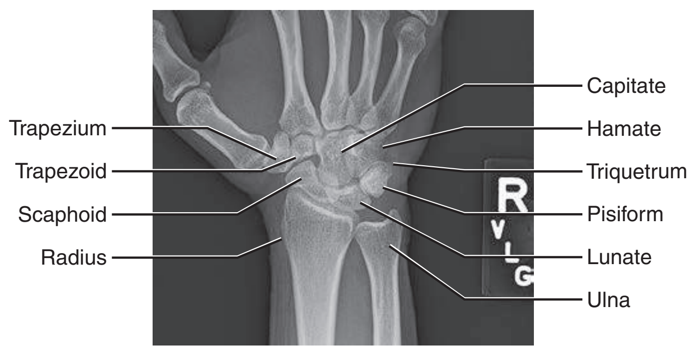


Fig. 4.93 PA oblique of right wrist.

Evaluation Criteria

Anatomy Demonstrated: • Distal radius, ulna, carpals, and at least to midmetacarpal area are visible. • Trapezium and scaphoid should be well visualized, with only slight superimposition of other carpals on their medial aspects (Figs. 4.92 and 4.93).
Position: • Long axis of the hand, wrist, and forearm should be aligned with IR. • True 45° oblique of the wrist is evidenced by: ulnar head partially superimposed by distal radius; proximal third through fifth metacarpals (metacarpal bases) should appear mostly superimposed. • CR and center of collimation field should be to **midcarpal area.**
Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate carpals and their overlapping borders; soft tissue margins; and clear, sharp bony trabecular markings.

LATEROMEDIAL PROJECTION—WRIST

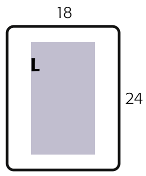
Clinical Indications

ROUTINE
- PA
- PA oblique
- Lateral

- Fractures or dislocations of the distal radius or ulna, specifically anteroposterior fragment displacements for **Barton, Colles,** or **Smith fractures**
- Osteoarthritis also may be demonstrated primarily in the trapezium and first CMC joint

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—60 to 70



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with arm and forearm resting on the table. Place wrist and hand on IR in thumb-up lateral position. Shoulder, elbow, and wrist should be on same horizontal plane.

Part Position

- Align and center hand and wrist to long axis of IR.
- Adjust hand and wrist into a **true lateral** position, with fingers comfortably extended (Fig. 4.94); if support is needed to prevent motion, use a radiolucent support block and sandbag, and place block against extended hand and fingers (Fig. 4.95).

CR

- CR perpendicular to IR, directed to **midcarpal area**

Recommended Collimation Collimate on four sides, including distal radius and ulna and metacarpal area.

Fig. 4.94 Part position—lateral wrist

Fig. 4.95 Patient position—lateral wrist with support.

Evaluation Criteria

Anatomy Demonstrated: • Distal radius and ulna, carpals, and at least the midmetacarpal area are visible.
Position: • Long axis of the hand, wrist, and forearm should be aligned with long axis of IR. • **True lateral** position is evidenced by: ulnar head should be superimposed over distal radius; proximal second through fifth metacarpals all should appear aligned and superimposed (Figs. 4.96 and 4.97). • CR and center of collimation field should be to **midcarpal region.**
Exposure: • Optimal density (brightness) and contrast with **no motion** demonstrate clear, sharp bony 'trabecular markings and soft tissue, such as margins of pertinent fat pads of the wrist and borders of the distal ulna, seen through the superimposed radius.

Fig. 4.96 Lateral projection of left wrist.

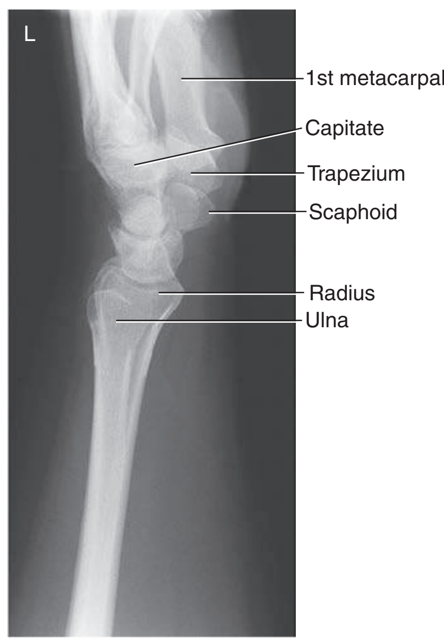


Fig. 4.97 Lateral wrist.

PA AND PA AXIAL SCAPHOID—WITH ULNAR DEVIATION: WRIST

WARNING: If patient has possible wrist trauma, do *not* attempt this position before a routine wrist series has been completed and evaluated to rule out possible fracture of distal forearm or wrist or both.

Wrist
SPECIAL
- Scaphoid projections: CR angle with ulnar deviation

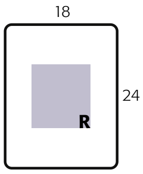
Clinical Indications

- Possible fractures of the **scaphoid**

Nondisplaced fractures may require additional projections or CT scan of the wrist

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with wrist and hand on IR, palm down, and shoulder, elbow, and wrist on same horizontal plane.

Part Position

- Position wrist as for a PA projection—palm down and hand and wrist aligned with center of long axis of IR, with scaphoid centered to CR.
- Without moving forearm, gently evert hand (move toward ulnar side) as far as patient can tolerate without lifting or rotating distal forearm (Fig. 4.98).

NOTE: See terminology in Chapter 1 for explanation of ulnar deviation versus radial deviation.

CR

- Angle CR **10° to 15° proximally,** along long axis of forearm and toward elbow (CR angle should be perpendicular to long axis of scaphoid).
- Center CR to **scaphoid** (Locate scaphoid at a point ¾ inch (2 cm) distal and medial to radial styloid process).

Recommended Collimation Collimate on four sides to carpal region.

NOTE: Obscure fractures of scaphoid may require several projections taken with different CR angles. Rafert and Long[11] described a four-projection series with the CR angled proximally 0°, 10°, 20°, and 30°.

4

Evaluation Criteria

Anatomy Demonstrated: • Distal radius and ulna, carpals, and proximal metacarpals are visible. • Scaphoid should be demonstrated clearly without foreshortening, with adjacent carpal interspaces open (evidence of CR angle) (Figs. 4.99 and 4.100).

Position: • Long axis of wrist and forearm should be aligned with side border of IR. • Ulnar deviation should be evident by the angle of the long axis of the metacarpals to that of the radius and ulna. • **No rotation** of wrist is evidenced by appearance of distal radius and ulna, with minimal superimposition of distal radioulnar joint (Fig. 4.101). • CR and center of collimation field should be to the **scaphoid.**

Exposure: • Optimal density (brightness) and contrast with **no motion** visualize the scaphoid borders and clear, sharp bony trabecular markings.

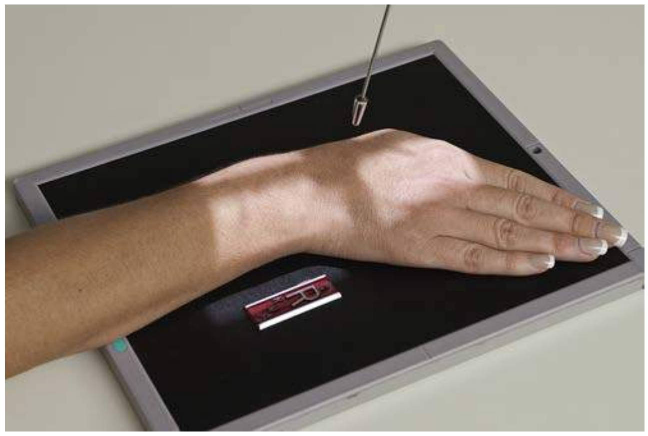


Fig. 4.98 PA axial wrist (scaphoid)—ulnar deviation with 15° CR angle.

Fig. 4.99 15° CR angle.

Fig. 4.100 25° CR angle.

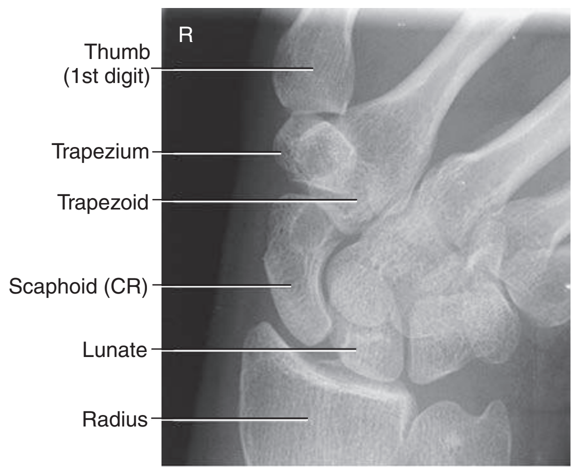


Fig. 4.101 15° CR angle.

PA SCAPHOID—HAND ELEVATED AND ULNAR DEVIATION: WRIST

MODIFIED STECHER METHOD[12]

WARNING: If patient has possible wrist trauma, do *not* attempt this position before a routine wrist series has been completed and evaluated to rule out possible fracture of distal forearm or wrist or both.

Clinical Indications

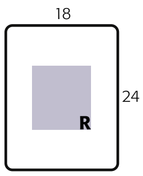
- Possible fractures of the **scaphoid**

This is an alternative projection to the CR angle ulnar deviation method demonstrated on the preceding page.

Wrist
ALTERNATE
- Scaphoid projections: CR angle with ulnar deviation
- Scaphoid projections: Hand elevated and ulnar deviation, modified Stecher method

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended. Drop shoulder so that shoulder, elbow, and wrist are on same horizontal plane.

Part Position

- Place hand and wrist palm down on IR with hand elevated on 20° angle sponge (Fig. 4.102).
- Ensure that wrist is in direct contact with IR.
- Gently evert or turn hand outward (toward ulnar side) unless contraindicated because of severe injury (Fig. 4.103).

Alternative Method Have patient clench the fist with ulnar deviation to obtain a similar position of the scaphoid.

CR

- Center CR **perpendicular to IR** and directed to **scaphoid** (locate scaphoid at a point ¾ inch [2 cm] distal and medial to radial styloid process).

Recommended Collimation Collimate on four sides to carpal region.

NOTE: Stecher[12] indicated that elevation of the hand 20° rather than angling of CR places the scaphoid parallel to the IR. Stecher also suggested that clenching of the fist is an alternative to elevation of the hand or angling of the CR. Bridgman[13] recommended ulnar deviation in addition to hand elevation, for less scaphoid superimposition.

Fig. 4.102 PA wrist for scaphoid as follows: hand elevated 20°; ulnar deviation, if possible; no CR angle.

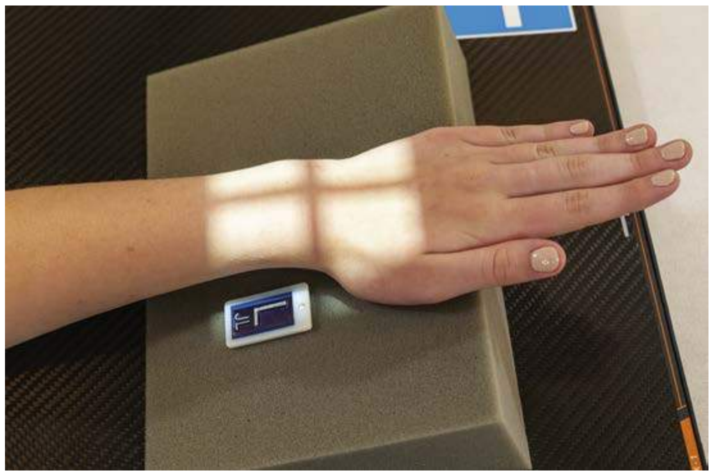


Fig. 4.103 Severe pain as follows: hand elevated 20°; **no** ulnar deviation; no CR angle.

Evaluation Criteria

Anatomy Demonstrated: • Distal radius and ulna, carpals, and proximal metacarpals are visible. • Carpals are visible, with adjacent interspaces more open on the lateral (radial) side of the wrist. • Scaphoid is shown, without foreshortening or superimposition of adjoining carpals (Figs. 4.104 and 4.105).

Position: • Long axis of wrist and forearm should be aligned with side border of IR. • Ulnar deviation is evidenced by only minimal, if any, superimposition of distal scaphoid. • **No rotation** of wrist is evidenced by the appearance of distal radius and ulna with no or only minimal superimposition of distal radioulnar joint. • CR and center of collimation field should be to **scaphoid.**

Exposure: • Optimal density (brightness) and contrast with **no motion** visualize the scaphoid borders and clear, sharp bony trabecular markings.

Fig. 4.104 Hand elevated, ulnar deviation, and no CR angle.

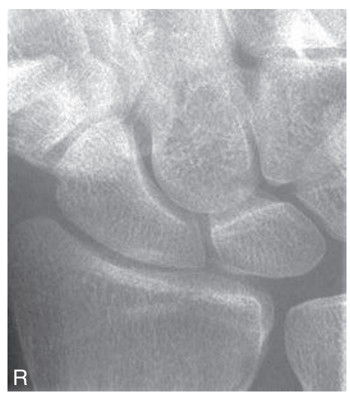


Fig. 4.105 Hand elevated, no ulnar deviation or CR angle.

PA PROJECTION—RADIAL DEVIATION: WRIST

WARNING: If patient has possible wrist trauma, do *not* attempt this position before a routine wrist series has been completed and evaluated to rule out possible fracture of distal forearm or wrist or both.

Wrist

SPECIAL

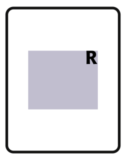
- Scaphoid projections: CR angle with ulnar deviation, or alternate modified Stecher method
- Radial deviation

Clinical Indications

- Possible fractures of the carpal bones on the ulnar side of the wrist, especially the lunate, triquetrum, pisiform, and hamate

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table with hand and forearm extended. Drop shoulder so that shoulder, elbow, and wrist are on same horizontal plane.

Part Position

- Position wrist as for PA projection—palm down with wrist and hand aligned with center of long axis of IR.
- Without moving forearm, gently invert the hand (move medially toward thumb side) as far as patient can tolerate without lifting or rotating distal forearm (Fig. 4.106).

CR

- CR perpendicular to IR, directed to midcarpal area

Recommended Collimation Collimate on four sides to carpal region.

Evaluation Criteria

Anatomy Demonstrated: • Distal radius and ulna, carpals, and proximal metacarpals are visible. • Carpals are visible, with adjacent interspaces more open on the medial (ulnar) side of the wrist (Figs. 4.107 and 4.108).

Position: • Long axis of the forearm is aligned with the side border of IR • Extreme radial deviation is evidenced by the angle of the long axis of the metacarpals to that of the radius and ulna and the space between the triquetrum/pisiform and the styloid process of the ulna. • **No rotation** of the wrist is evidenced by the appearance of the distal radius and ulna. • CR and center of the collimation field should be to the **midcarpal area.**

Exposure: • Optimal density (brightness) and contrast with **no motion** visualize the carpal borders and clear, sharp bony trabecular markings.

Fig. 4.106 PA wrist—radial deviation.

Fig. 4.107 Radial deviation.

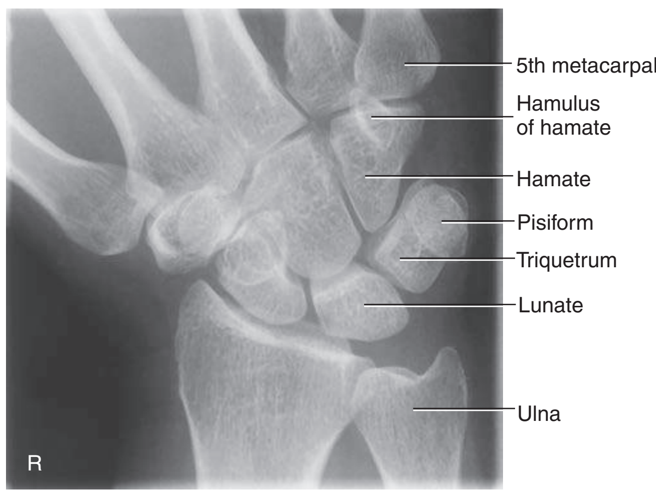


Fig. 4.108 Radial deviation.

CARPAL CANAL (TUNNEL)—TANGENTIAL, INFEROSUPERIOR PROJECTION: WRIST

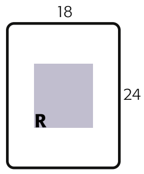
GAYNOR-HART METHOD

WARNING: If patient has possible wrist trauma, do *not* attempt this position before a routine wrist series has been completed and evaluated to rule out possible fracture of distal forearm or wrist or both.

Clinical Indications
- Rule out abnormal calcification and bony changes in the carpal sulcus that may impinge on the **median nerve,** as with **carpal tunnel syndrome**
- Possible fractures of the hamulus process of the hamate, pisiform, and trapezium

> **Wrist**
> SPECIAL
> - Scaphoid projections: CR angle with ulnar deviation, or alternate modified Stecher method
> - Radial deviation
> - Carpal canal

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with wrist and hand on IR and palm down (pronated).

Part Position
- Align hand and wrist with long axis of the IR.
- Ask patient to hyperextend wrist (dorsiflex) as far as possible by the use of a piece of tape or band, gently but firmly hyperextending the wrist until the long axis of the metacarpals and the fingers are as near vertical (90° to forearm) as possible (without lifting the wrist and forearm from the IR).
- Rotate entire hand and wrist about **10° internally** (toward radial side) to prevent superimposition of pisiform and hamate (Fig. 4.109).

CR
- Angle CR **25° to 30° proximally, to the long axis of the hand** (the total CR angle in relationship to the IR must be increased if patient cannot hyperextend wrist as far as indicated).
- Direct CR to a point about 1 inch (2 to 3 cm) distal to the base of third metacarpal (center of palm of hand).

Recommended Collimation Collimate on four sides to area of interest.

Alternative Imaging Sonography for carpal tunnel: High-resolution ultrasonography allows for noninvasive imaging of the carpal tunnel and related anatomy. Fig. 4.110 demonstrates the "bowing" of the flexor retinaculum (arrows) with the "flattening" of the median nerve below it, indicating compression.[14]

Fig. 4.109 Tangential projection. CR 25° to 30° to long axis of hand.

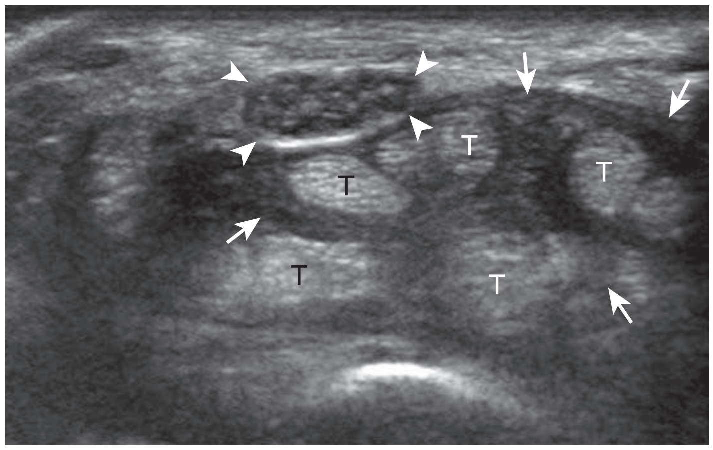


Fig. 4.110 Carpal tunnel syndrome: tenosynovitis. Ultrasound image in short axis. (From Jacobson J. *Fundamentals of musculoskeletal ultrasound,* ed 2, Philadelphia, 2013, Elsevier.)

Evaluation Criteria

Anatomy Demonstrated: • The carpals are demonstrated in a tunnel-like, arched arrangement (Figs. 4.111 and 4.112).

Position: • The pisiform and the hamulus process should be separated and visible in profile without superimposition. • The rounded palmar aspects of the capitate and the scaphoid should be visualized in profile, in addition to the aspect of the trapezium that articulates with the first metacarpal. • CR and center of collimation field should be to **midpoint of the carpal canal.**

Exposure: • Optimal density (brightness) and contrast should visualize soft tissues and possible calcifications in carpal canal region, and outlines of superimposed carpals should be visible without overexposure of these carpals in profile. • Trabecular markings and bony margins should appear clear and sharp, indicating **no motion.**

Fig. 4.111 Tangential (Gaynor-Hart method) projection.

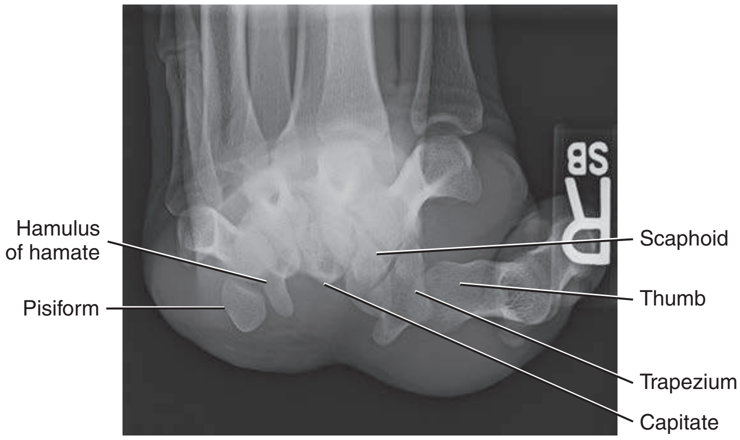


Fig. 4.112 Tangential projection of right wrist.

CARPAL BRIDGE—TANGENTIAL PROJECTION: WRIST

WARNING: If patient has possible wrist trauma, do *not* attempt this position before a routine wrist series has been completed and evaluated to rule out possible fracture of distal forearm or wrist or both.

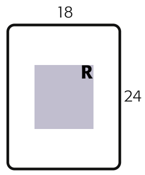
Wrist
SPECIAL
- Scaphoid projections: CR angle with ulnar deviation, alternate modified Stecher method
- Radial deviation
- Carpal canal
- Carpal bridge

Clinical Indications

- Calcification or other pathology of the dorsal (posterior) aspect of the carpal bones

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—55 to 65



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Have patient stand or sit at end of the table and then lean over and place dorsal surface of hand, **palm upward,** on IR.

Part Position

- Center dorsal aspect of carpals to IR.
- Gently flex wrist as far as patient can tolerate, or until the hand and forearm form as near a 90° (right) angle as possible (Fig. 4.113).

CR

- Angle CR 45°distally, to the long axis of the forearm.
- Direct CR to a midpoint of the distal forearm about 1½ inches (4 cm) proximal to wrist joint.

Recommended Collimation Collimate all four sides of carpal region.

Evaluation Criteria

Anatomy Demonstrated: • Tangential view of the dorsal aspect of the scaphoid, lunate, and triquetrum is visible. • Outline of the capitate and trapezium superimposed is visible (Figs. 4.114 and 4.115).

Position: • Dorsal aspect of the carpal bones should be visualized clear of superimposition and centered to IR. • CR and center of collimation field should be to the area of the **dorsal carpal bones.**

Exposure: • Optimal density (brightness) and contrast with **no motion** should demonstrate the dorsal aspect of carpal bones, with sharp borders and clear, sharp bony trabecular markings. • Outlines of proximal metacarpals should be visualized through superimposed structures without overexposure of the dorsal aspects of carpals seen in profile.

Fig. 4.113 Carpal bridge—tangential projection; CR 45° to forearm.

Fig. 4.114 Carpal bridge tangential projection of right wrist.

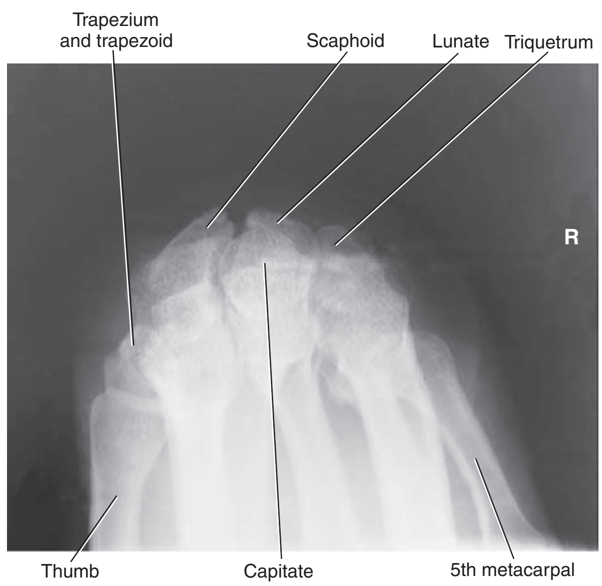


Fig. 4.115 Carpal bridge.

4

AP PROJECTION—FOREARM

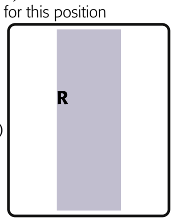
Clinical Indications

- Fractures and dislocations of the radius or ulna
- Pathologic processes such as osteomyelitis or arthritis

Forearm
ROUTINE
• AP
• Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size 14 × 17 inches (35 × 43 cm), portrait or smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with hand and arm fully extended and **palm up (supinated).**

Part Position

- Drop shoulder to place entire upper limb on same horizontal plane.
- Align and center forearm to long axis of IR, ensuring that both wrist and elbow joints are included (use as large an IR as necessary).
- Instruct patient to lean laterally as necessary to place entire wrist, forearm, and elbow in as near a true frontal position as possible (Fig. 4.116). Palpate the medial and lateral epicondyles to ensure they are the same distance from IR.

CR

- CR perpendicular to IR, directed to **mid-forearm**

Recommended Collimation Collimate lateral borders to actual forearm area with minimal collimation at both ends to avoid excluding anatomy at either joint. Considering divergence of the x-ray beam, ensure that a **minimum** of 1 to 1½ inches (3 to 4 cm) distal to wrist and elbow joints is included on IR.

Evaluation Criteria

Anatomy Demonstrated: • AP projection of the entire radius and ulna is shown, with a minimum of proximal row carpals and distal humerus and pertinent soft tissues, such as fat pads and stripes of the wrist and elbow joints (Fig. 4.117).
Position: • Long axis of forearm should be aligned with long axis of IR. • **No rotation** is evidenced by humeral epicondyles visualized in profile, with radial head, neck, and tuberosity slightly superimposed by the ulna. • Wrist and elbow joint spaces are only partially open because of beam divergence. • CR and center of collimation field should be to the **approximate midpoint of the radius and ulna.**
Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize soft tissue and sharp, cortical margins and clear, bony trabecular markings.

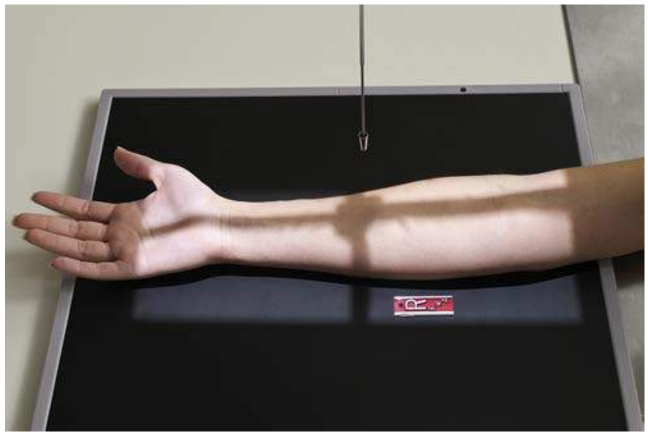


Fig. 4.116 AP forearm (including both joints).

Fig. 4.117 AP forearm (both joints).

4

LATEROMEDIAL PROJECTION—FOREARM

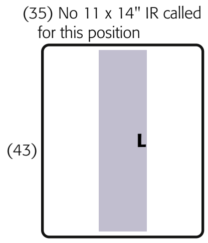
Forearm
ROUTINE
• AP
• Lateral

Clinical Indications

- Fractures and dislocations of the radius or ulna
- Pathologic processes, such as osteomyelitis or arthritis

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75
- To make best use of the anode heel effect, place elbow at cathode end of x-ray beam



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow flexed 90°.

Part Position

- Drop shoulder to place entire upper limb on same horizontal plane.
- Align and center forearm to long axis of IR; ensure that both wrist and elbow joints are included on IR (Fig. 4.118).
- Rotate hand and wrist into **true lateral position,** and support hand to prevent motion, if needed (ensure that distal radius and ulna are superimposed directly).
- For heavy muscular forearms, place support under hand and wrist as needed to place radius and ulna parallel to IR.

CR

- CR perpendicular to IR, directed to **mid-forearm**

Recommended Collimation Collimate both lateral borders to the actual forearm area. Also, collimate at both ends to avoid excluding anatomy at either joint. Considering divergence of the x-ray beam, ensure that a **minimum** of 1 to 1½ inches (3 to 4 cm) distal to wrist and elbow joints is included on IR.

Evaluation Criteria

Anatomy Demonstrated: • Lateral projection of entire radius and ulna, proximal row of carpal bones, elbow, and distal end of the humerus are visible, in addition to pertinent soft tissue, such as fat pads and stripes of the wrist and elbow joints (Fig. 4.119).

Position: • Long axis of forearm should be aligned with long axis of IR. • Elbow should be flexed 90°. • **No rotation** as evidenced by head of ulna being superimposed over the radius, and humeral epicondyles should be superimposed. • Radial head should superimpose coronoid process, with radial tuberosity demonstrated. • CR and center of collimation field should be to **midpoint of the radius and ulna.**

Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize sharp cortical margins and clear, sharp bony trabecular markings and fat pads and stripes of the wrist and elbow joints.

Fig. 4.118 Lateral forearm (including both joints).

Fig. 4.119 Lateral projection of forearm (both joints).

AP PROJECTION—ELBOW

ELBOW FULLY EXTENDED

Elbow

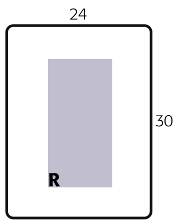
ROUTINE
- AP
- Alternate AP—partial flexion
- Alternate AP—acute flexion
- Oblique
- Lateral (external)
- Medial (internal)
- Lateral

Clinical Indications

- Fractures and dislocations of the elbow
- Pathologic processes, such as osteomyelitis and arthritis

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow fully extended, if possible (See following page if patient cannot fully extend elbow).

Part Position

- Extend elbow, supinate hand, and align arm and forearm with long axis of IR (Fig. 4.120).
- Center elbow joint to center of IR.
- Ask patient to lean laterally as necessary for **true AP projection.** Palpate humeral epicondyles to ensure that the interepicondylar plane is parallel to the IR. (The interepicondylar plane is an imaginary plane between the medial and lateral epicondyles of the distal humerus. This plane is useful for elbow and humerus positioning.)
- Support hand as needed to prevent motion.

CR

- CR perpendicular to IR, directed to **mid-elbow joint,** which is approximately ¾ inch (2 cm) distal to midpoint of a line between epicondyles

Recommended Collimation Collimate on four sides to area of interest.

Evaluation Criteria

Anatomy Demonstrated: • Distal humerus, elbow joint space, and proximal radius and ulna are visible (Figs. 4.121 and 4.122).
Position: • Long axis of arm should be aligned with long axis of IR. • **No rotation** is evidenced by the appearance of bilateral epicondyles seen in profile and radial head, neck, and tubercles separated or only slightly superimposed by ulna. • Olecranon process should be seated in the olecranon fossa with fully extended arm • Elbow joint space appears open with fully extended arm and proper CR centering. • CR and center of collimation field should be to the **mid-elbow joint.**
Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize soft tissue detail; sharp, bony cortical margins; and clear, bony trabecular markings.

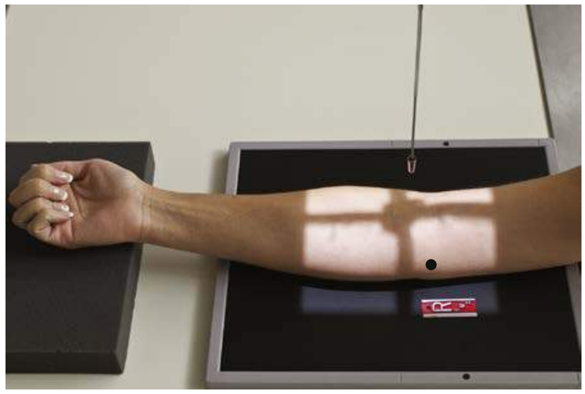


Fig. 4.120 AP elbow (fully extended).

Fig. 4.121 AP (extended).

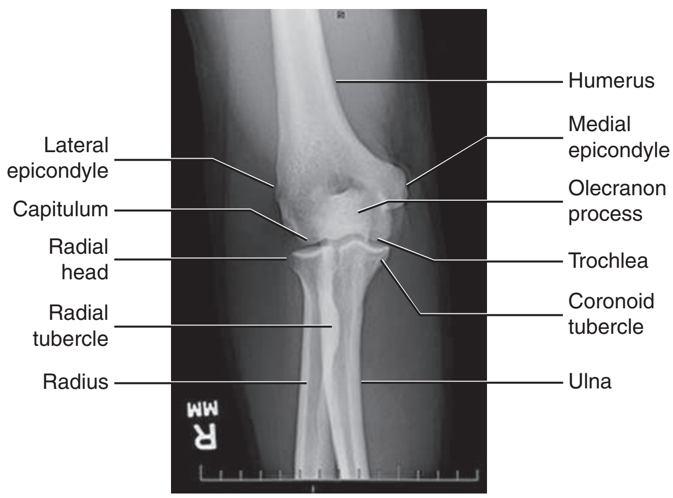


Fig. 4.122 AP right elbow (extended).

4

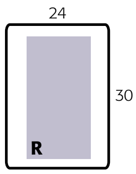
AP PROJECTION—ALTERNATE PARTIAL FLEXION: ELBOW

WHEN ELBOW CANNOT BE FULLY EXTENDED

Clinical Indications
- Fractures and dislocations of the elbow
- Pathologic processes, such as osteomyelitis and arthritis

Elbow
ROUTINE
- AP
- Alternate AP—partial flexion
- Alternate AP—acute flexion
- Oblique
- Lateral (external)
- Medial (internal)
- Lateral

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow partially flexed.

Part Position 
- Obtain **two** AP projections—one with **forearm parallel** to IR and one with **humerus parallel** to IR (Figs. 4.123 and 4.124).
- Place support under wrist and forearm for projection with humerus parallel to IR, if needed, to prevent motion.

CR
- CR **perpendicular** to IR, directed to **mid-elbow joint,** which is approximately ¾ inch (2 cm) distal to midpoint of a line between epicondyles

Recommended Collimation Collimate on four sides to area of interest.

NOTE: If patient cannot partially extend elbow as shown (see Fig. 4.123) and elbow remains **flexed near 90°,** take the two AP projections as described, but **angle CR 10° to 15°** into elbow joint, or if flexed **more than 90°,** use the **acute flexion projection** (see p. 166).

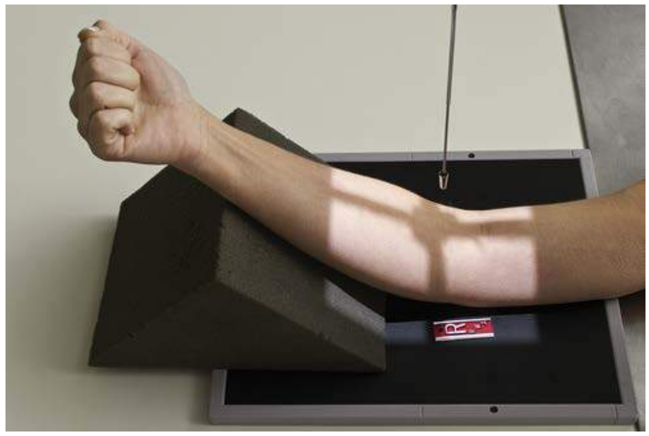


Fig. 4.123 AP elbow (partially flexed); humerus parallel to IR.

Fig. 4.124 AP elbow (partially flexed); forearm parallel to IR.

Evaluation Criteria

Anatomy Demonstrated: • Distal humerus is best visualized on "humerus parallel" projection, and proximal radius and ulna are best visualized on "forearm parallel" projection (Figs. 4.125 and 4.126).

NOTE: Structures in elbow joint region are partially obscured and slightly distorted, depending on amount of elbow flexion possible.

Position: • Long axis of arm should be aligned with side border of IR. • **No rotation** is evidenced by the epicondyles seen in profile and radial head and neck separated or only slightly superimposed over ulna on forearm parallel projection. • CR and center of collimation field should be to the **mid-elbow joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize soft tissue detail; sharp, bony cortical margins; and clear, bony trabecular markings. • Distal humerus, including epicondyles, should be demonstrated with sufficient density on "humerus parallel" projection. • On "forearm parallel" projection, proximal radius and ulna should be well visualized with density to allow visualization of both soft tissue and bony detail.

Fig. 4.125 Humerus parallel.

Fig. 4.126 Forearm parallel.

ACUTE FLEXION PROJECTIONS—ELBOW

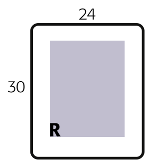
AP PROJECTIONS OF ELBOW IN ACUTE FLEXION

Elbow
ROUTINE
- AP
- Alternate AP—partial flexion
- Alternate AP—acute flexion
- Oblique
- Lateral (external)
- Medial (internal)
- Lateral

Clinical Indications

- Fractures and moderate dislocations of the elbow in acute flexion when the elbow cannot be extended to any degree

NOTE: To visualize both the distal humerus and the proximal radius and ulna, **two** projections are required—one with **CR perpendicular to the humerus** and one with CR angled so that it is **perpendicular to the forearm.**



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—70 to 80

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with acutely flexed arm resting on IR.

Part Position

- Align and center humerus to long axis of IR, with forearm acutely flexed and fingertips resting on shoulder.
- Adjust IR to center of elbow joint region.
- Palpate humeral epicondyles and ensure interepicondylar plane is parallel to IR for **no rotation**

CR

- **Distal humerus:** CR perpendicular to IR and humerus, directed to a point midway between epicondyles (Fig. 4.127)
- **Proximal forearm:** CR perpendicular to forearm (angling CR as needed), directed to a point approximately 2 inches (5 cm) proximal or superior to olecranon process (Fig. 4.128)

Recommended Collimation Collimate on four sides to area of interest.

Evaluation Criteria

- Four-sided collimation borders should be visible with CR and center of collimation field midway between epicondyles.

Distal Humerus: • Forearm and humerus should be directly superimposed. • Medial and lateral epicondyles and parts of trochlea, capitulum, and olecranon process all should be seen in profile. • Optimal exposure should visualize distal humerus and olecranon process through superimposed structures. • Soft tissue detail is not readily visible on either projection (Figs. 4.129 and 4.131).

Proximal Forearm: • Proximal ulna and radius, including outline of radial head and neck, should be visible through superimposed distal humerus. • Optimal exposure visualizes outlines of proximal ulna and radius superimposed over humerus (Figs. 4.130 and 4.132).

Fig. 4.127 For distal humerus—CR perpendicular to **humerus.**

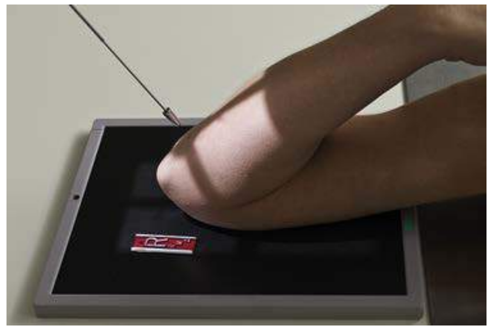


Fig. 4.128 For proximal forearm—CR perpendicular to **forearm.**

Fig. 4.129 Distal humerus.

Fig. 4.130 Proximal forearm.

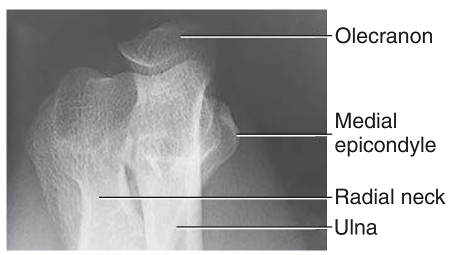


Fig. 4.131 Distal humerus.

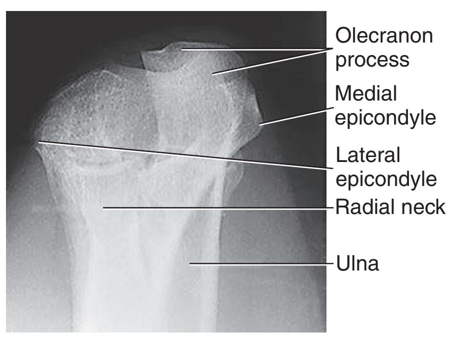


Fig. 4.132 Proximal forearm.

AP OBLIQUE PROJECTION—LATERAL (EXTERNAL) ROTATION: ELBOW

Clinical Indications

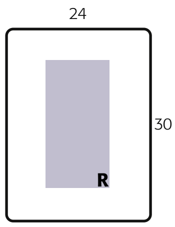
- Fractures and dislocations of the elbow, primarily the radial head and neck
- Certain pathologic processes, such as osteomyelitis and arthritis

Lateral (External Rotation) Oblique Best visualizes radial head and neck of the radius and capitulum of humerus

Elbow
ROUTINE
- AP
- Alternate AP—partial flexion
- Alternate AP—acute flexion
- Oblique
- Lateral (external)
- Medial (internal)
- Lateral

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range —65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with arm fully extended and shoulder and elbow on same horizontal plane (lowering shoulder as needed).

Part Position

- Align arm and forearm with long axis of IR (Fig. 4.133).
- Center elbow joint to CR and to IR.
- Supinate hand and **rotate laterally** the entire arm so that the distal humerus and the anterior surface of the elbow joint are approximately 45° to IR. (Patient must lean laterally for sufficient lateral rotation.) Place interepicondylar plane approximately 45° to the IR (Fig. 4.134).

CR

- CR perpendicular to IR, directed to **mid-elbow joint** (a point approximately ¾ inch [2 cm] distal to midpoint of line between the epicondyles as viewed from the x-ray tube)

Recommended Collimation Collimate on four sides to area of interest.

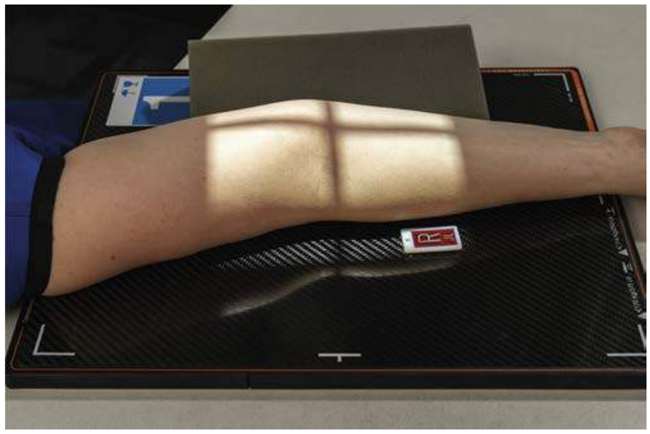


Fig. 4.133 45° lateral (external) oblique.

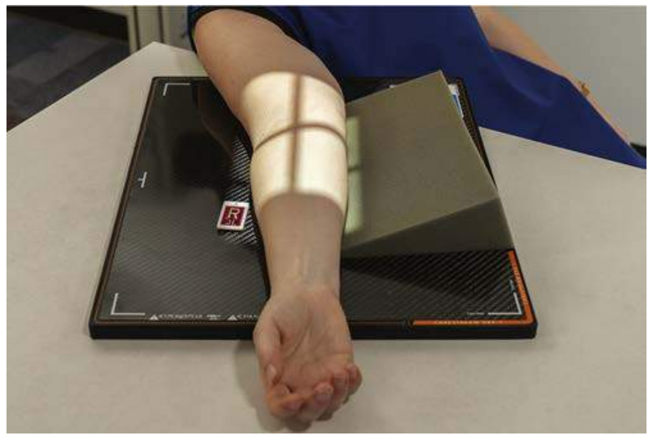


Fig. 4.134 End view, showing 45° external rotation.

4

Evaluation Criteria

Anatomy Demonstrated: • Oblique projection of distal humerus and proximal radius and ulna is visible (Figs. 4.135 and 4.136).

Position: • Long axis of arm should be aligned with side border of IR. • Correct 45° lateral oblique should visualize **radial head, neck,** and **tuberosity,** free of superimposition by ulna. • Lateral epicondyle and capitulum should appear elongated and in profile. • CR and center of collimation field should be to **mid-elbow joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize soft tissue detail; sharp, bony cortical margins; and clear, bony trabecular markings.

Fig. 4.135 Lateral oblique of right elbow—external rotation.

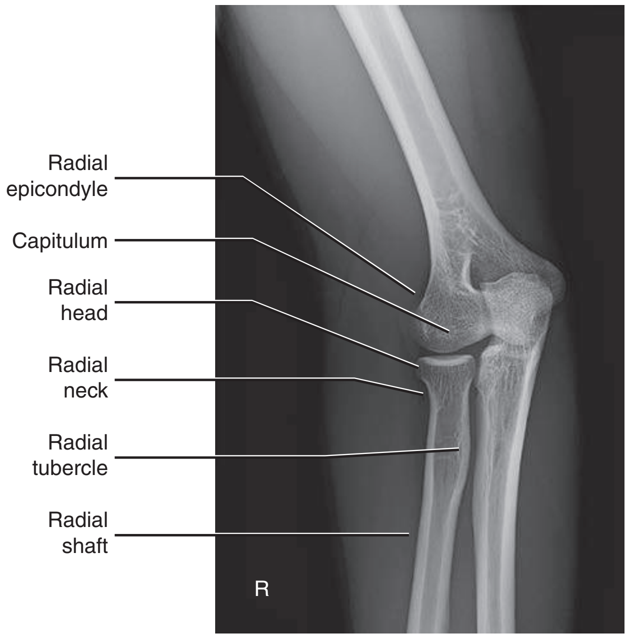


Fig. 4.136 Lateral oblique—external rotation of right elbow.

AP OBLIQUE PROJECTION—MEDIAL (INTERNAL) ROTATION: ELBOW

Elbow
ROUTINE
- AP
- Alternate AP—partial flexion
- Alternate AP—acute flexion
- Oblique
- Lateral (external)
- Medial (internal)
- Lateral

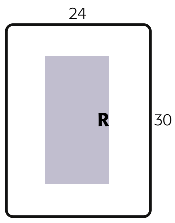
Clinical Indications

- Fractures and dislocations of the elbow, primarily the coronoid process
- Certain pathologic processes, such as osteoporosis and arthritis

Medial (Internal Rotation) Oblique Best visualizes **coronoid process** of ulna and **trochlea** in profile.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with arm fully extended and shoulder and elbow on same horizontal plane.

Part Position

- Align arm and forearm with long axis of IR. Center elbow joint to CR and to IR.
- Pronate hand into a natural palm-down position and rotate arm as needed until distal humerus and anterior surface of elbow are rotated **45°** (place interepicondylar plane approximately 45° to the IR) (Figs. 4.137 and 4.138).

CR

- CR perpendicular to IR, directed to **mid-elbow joint** (approximately ¾ inch [2 cm] distal to midpoint of line between epicondyles as viewed from x-ray tube)

Recommended Collimation Collimate on four sides to area of interest.

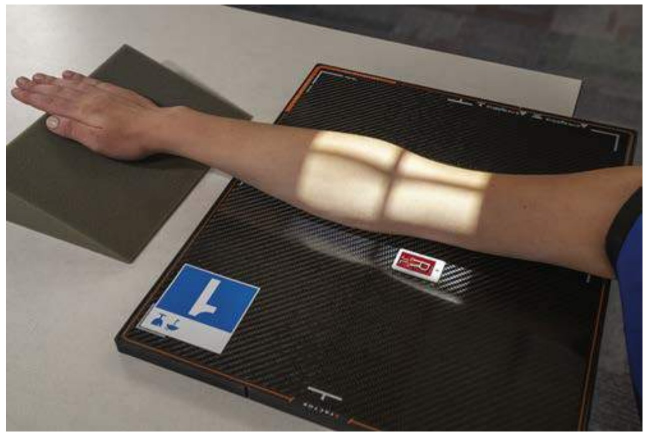


Fig. 4.137 Medial (internal rotation) oblique.

Fig. 4.138 End view, showing 45° medial oblique.

Fig. 4.139 Medial (internal rotation) oblique.

Evaluation Criteria

Anatomy Demonstrated: • Oblique view of distal humerus and proximal radius and ulna is visible (Figs. 4.139 and 4.140).

Position: • Long axis of arm should be aligned with side border of IR. • Correct 45° medial oblique should visualize coronoid process of the ulna in profile. • Radial head and neck should be superimposed and centered over the proximal ulna. • Medial epicondyle and trochlea should appear elongated and in partial profile. • Olecranon process should appear seated in olecranon fossa and trochlear notch partially open and visualized with arm fully extended. • CR and center of collimation field should be at **mid-elbow joint.**

Exposure: • Optimal density (brightness) and contrast with **no motion** should visualize soft tissue detail; bony cortical margins; and clear, bony trabecular markings.

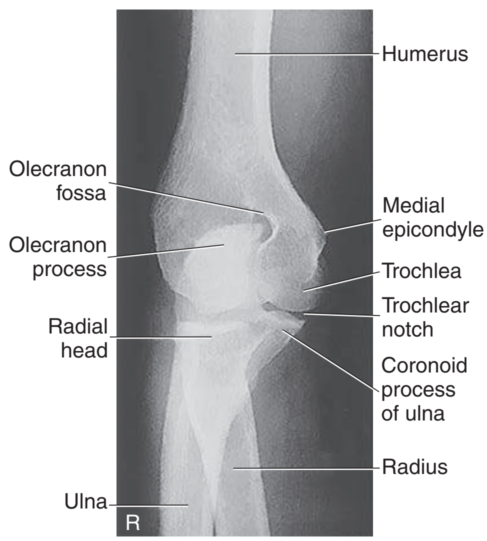


Fig. 4.140 Medial oblique of right elbow.

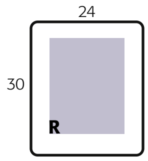
4

LATEROMEDIAL PROJECTION—ELBOW

Elbow
ROUTINE
- AP
- Alternate AP—partial flexion
- Alternate AP—acute flexion
- Oblique
- Lateral (external)
- Medial (internal)
- Lateral

Clinical Indications
- Fractures and dislocations of the elbow
- Certain bony pathologic processes, such as osteomyelitis and arthritis
- Elevated or displaced fat pads of the elbow joint may be visualized

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with elbow flexed 90° (see NOTE).

Part Position
- Align long axis of forearm with long axis of IR.
- Center elbow joint to CR and to center of IR.
- Drop shoulder so that humerus and forearm are on same horizontal plane.
- Rotate hand and wrist into true lateral position, thumb side up. Place interepicondylar plane perpendicular to the IR (Fig. 4.141).
- Place support under hand and wrist to elevate hand and distal forearm as needed for heavy muscular forearm so that forearm is parallel to IR for true lateral elbow.

CR
- CR perpendicular to IR, directed to **mid-elbow joint** (a point approximately 1½ inches [4 cm] medial to easily palpated posterior surface of olecranon process)

Recommended Collimation Collimate on four sides to area of interest.

NOTE: Diagnosis of certain important joint pathologic processes (e.g., possible visualization of the posterior fat pad) depends on 90° flexion of the elbow joint.[3]

EXCEPTION: Certain soft tissue diagnoses may require less flexion (30° to 35°), but these views should be taken only when specifically indicated.

4

Evaluation Criteria
Anatomy Demonstrated: • Lateral projection of distal humerus and proximal forearm, olecranon process, and soft tissues and fat pads of the elbow joint are visible (Figs. 4.142 and 4.143).
Position: • Long axis of the forearm should be aligned with long axis of IR, with the elbow joint flexed 90°. • About one-half of radial head should be superimposed by the coronoid process, and olecranon process should be visualized in profile. • True lateral view is indicated by three concentric arcs of the trochlear sulcus, double ridges of the capitulum and trochlea, and the trochlear notch of the ulna. In addition, superimposition of the humeral epicondyles occurs. • CR and center of collimation field should be **midpoint of the elbow joint.**
Exposure: • **No motion** and optimal density (brightness) and contrast should visualize sharp cortical margins and clear trabecular markings as well as soft tissue margins of the anterior and posterior fat pads.

Fig. 4.141 Lateral—elbow flexed 90° (forearm parallel to IR).

Fig. 4.142 Lateromedial projection of right elbow.

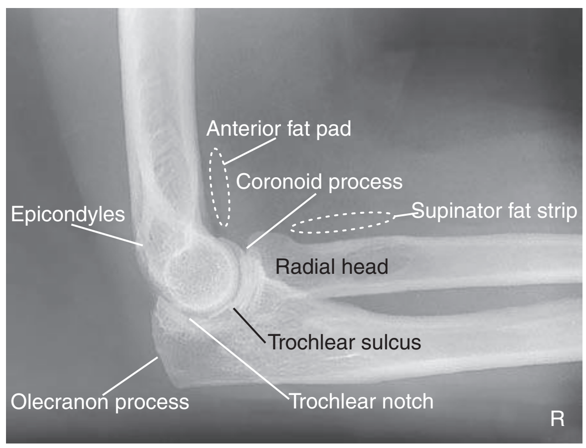


Fig. 4.143 Lateromedial projection of right elbow.

TRAUMA AXIAL LATEROMEDIAL AND MEDIOLATERAL PROJECTIONS: ELBOW

COYLE METHOD[15]

These are special projections taken for pathologic processes or trauma to the area of the radial head or the coronoid process of ulna. These are effective projections when the patient cannot extend the elbow fully for medial or lateral oblique projections of the elbow.

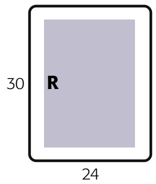
Elbow
SPECIAL
- Trauma axial laterals (Coyle method)

Clinical Indications

- Fractures and dislocations of the elbow, particularly the radial head (part position 1) and coronoid process (part position 2 for coronoid process)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—70 to 80



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at the end of the table for the erect position or supine on the table for horizontal beam imaging.

Part Position 1: Radial Head–Axial Lateromedial Projection

- Elbow flexed 90° if possible; **hand pronated**
- CR directed at **45° angle toward shoulder,** centered to radial head, mid-elbow joint (Figs. 4.144 and 4.146)

Part Position 2: Coronoid Process–Axial Mediolateral Projection

- Elbow flexed **only 80°** from extended position (because >80° may obscure coronoid process) and hand pronated
- CR angled **45° from shoulder,** into mid-elbow joint (Figs. 4.145 and 4.147)

Recommended Collimation Collimate on four sides to area of interest.

NOTE: Increase exposure factors by 4 to 6 kVp from lateral elbow because of angled CR. These projections are effective with or without a splint.

Evaluation Criteria

Radial Head: • Joint space between radial head and capitulum should be open and clear. • Radial head, neck, and tuberosity should be in profile and free of superimposition except for a small part of the coronoid process. • Distal humerus and epicondyles appear distorted because of 45° angle (Figs. 4.148 and 4.150).

Coronoid Process: • Anterior portion of the coronoid appears elongated but in profile. • Joint space between coronoid process and trochlea should be open and clear. • Radial head and neck should be superimposed by ulna. • Optimal exposure factors should visualize clearly the coronoid process in profile. Bony margins of superimposed radial head and neck should be visualized faintly through proximal ulna (Figs. 4.149 and 4.151).

Fig. 4.144 Erect for **radial head**—flexed **90°.**

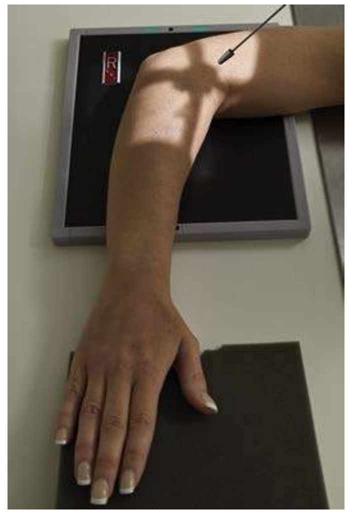


Fig. 4.145 Erect for **coronoid process**—flexed **80°.**

Fig. 4.146 Supine, angled 45° for **radial head**—flexed **90°.**

Fig. 4.147 Supine, angled 45° for **coronoid process**—flexed **80°.**

Fig. 4.148 For radial head.

Fig. 4.149 For coronoid process.

Fig. 4.150 Axial lateromedial projection for radial head.

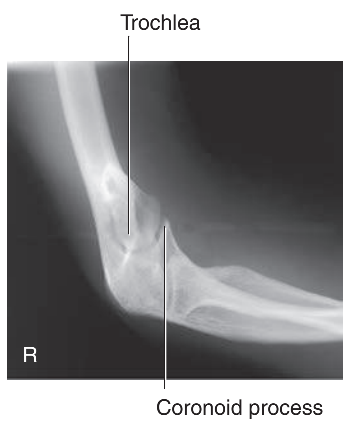


Fig. 4.151 Axial mediolateral projection for coronoid process.

RADIAL HEAD—LATEROMEDIAL PROJECTIONS: ELBOW

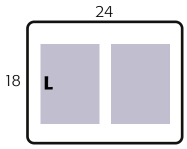
Clinical Indications

- Occult fractures of the radial head or neck

Elbow
SPECIAL
- Trauma axial laterals (Coyle method)
- Radial head laterals

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape; smallest IR available and collimate to area of interest
- Nongrid
- kVp range—65 to 75



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Seat patient at end of table, with arm **flexed 90°** and resting on IR with humerus, forearm, and hand on same horizontal plane. Place support under hand and wrist if needed.

Part Position

- Center radial head area to center of IR, positioned so that distal humerus and proximal forearm are placed "square" with, or parallel with, the borders of IR.
- Center radial head region to CR.
- Take **four projections,** the only difference among the four being rotation of the hand and wrist from (1) maximum external rotation to (4) maximum internal rotation; different parts of the radial head projected clear of the coronoid process are demonstrated. Near-complete rotation of the radial head occurs in these four projections, as follows:
  1. Supinate hand (palm up) and externally rotate as far as patient can tolerate (Fig. 4.152).
  2. Place hand in true lateral position (thumb up) (Fig. 4.154).
  3. Pronate hand (palm down) (Fig. 4.156).
  4. Internally rotate hand (thumb down) as far as patient can tolerate (Fig. 4.158).

CR

- CR perpendicular to IR, directed to **radial head** (approximately 1 inch [2 to 3 cm] distal to lateral epicondyle)

Recommended Collimation

- Collimate on four sides to area of interest (including at least 3 to 4 inches [10 cm] of proximal forearm and distal portion of humerus).

4

Evaluation Criteria

• Elbow should be flexed 90° in true lateral position, as evidenced by direct superimposition of epicondyles. • Radial head and neck should be partially superimposed by ulna but completely visualized in profile in various projections. • **Radial tuberosity** should be visualized in various positions and degrees of profile as follows (see small arrows): (1) Fig. 4.153, slightly anterior; (2) Fig. 4.155, not in profile, superimposed over radial shaft; (3) Fig. 4.157, slightly posterior; (4) Fig. 4.159, seen posteriorly, adjacent to ulna when hand and wrist are at maximum internal rotation. • Optimal exposure with **no motion** should clearly visualize sharp, bony margins and clear trabecular markings of radial head and neck area.


Fig. 4.152 1. Hand supinated (maximum external rotation).

Fig. 4.153 Hand supinated (maximum external rotation).


Fig. 4.154 2. Hand lateral.

Fig. 4.155 Hand lateral.


Fig. 4.156 3. Hand pronated.

Fig. 4.157 Hand pronated.


Fig. 4.158 4. Hand with maximum internal rotation.

Fig. 4.159 Hand with maximum internal rotation.

RADIOGRAPHS FOR CRITIQUE

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figures C4.160 through C4.165. Compare Image A to the other projections and identify the errors. While examining each image, consider the follow ing questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?
3. Are technical factors optimal?
4. Is there evidence of collimation and are pre-exposure anatomic side markers visible on the image?
5. Do these errors require a repeat exposure?

Feedback for each set of images is located on the faculty Evolve site.

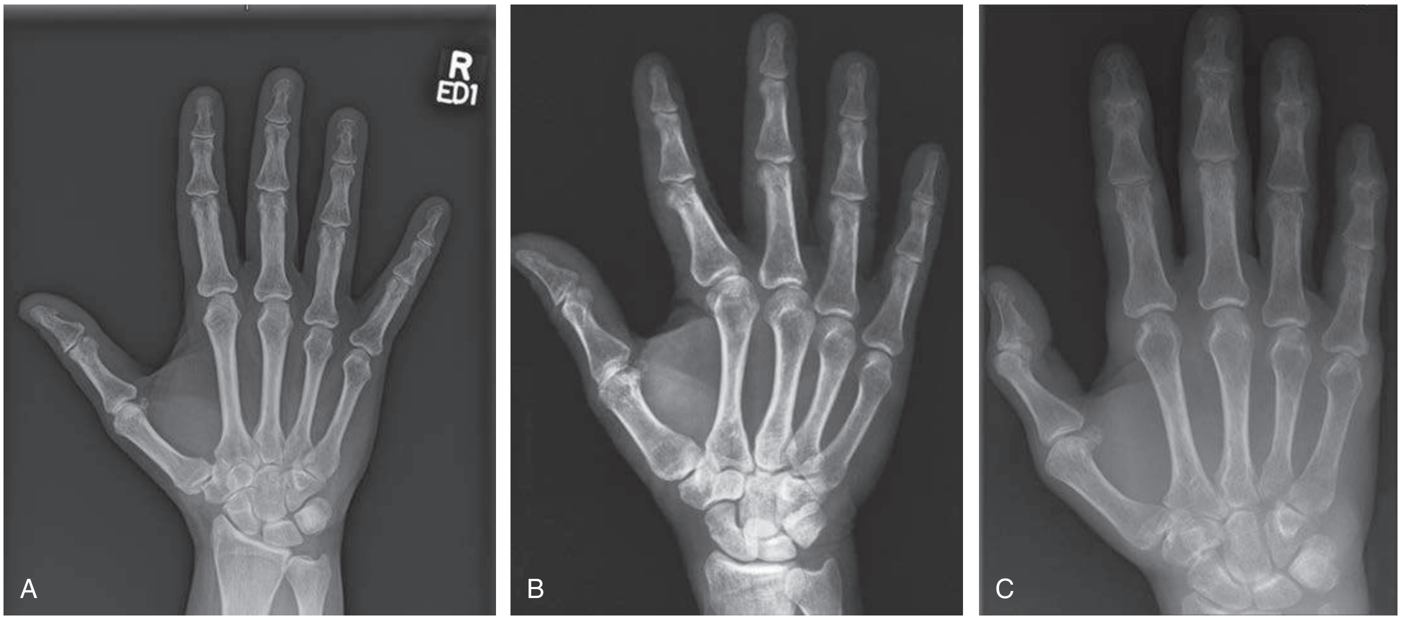


Fig. C4.160 PA hand.

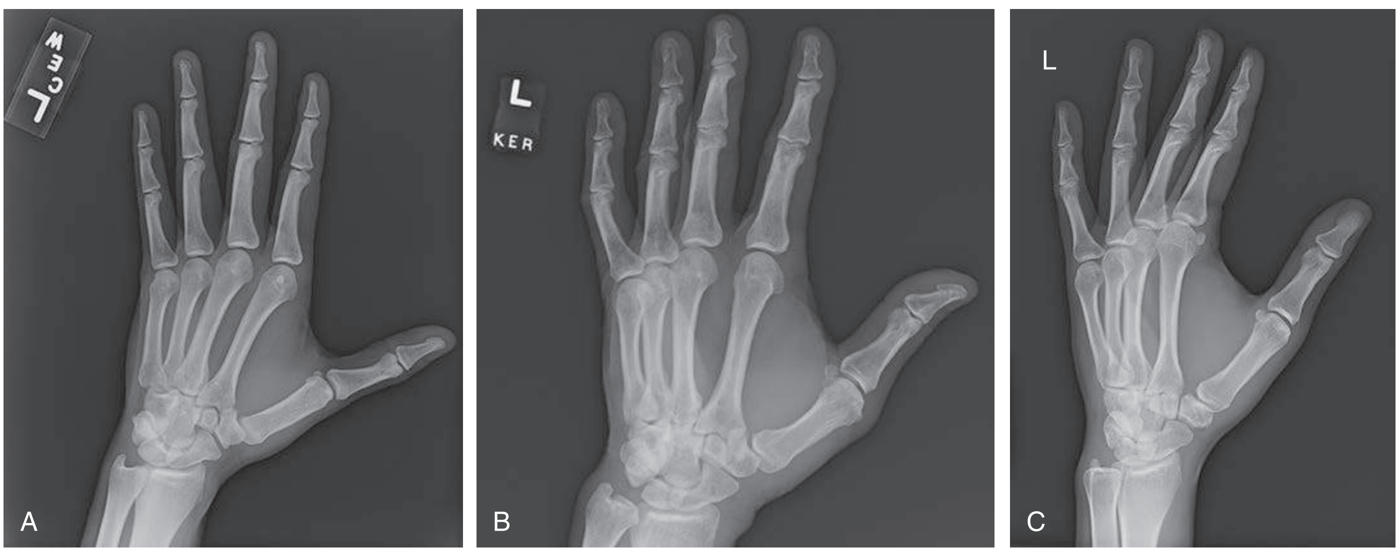


Fig. C4.161 PA oblique hand.

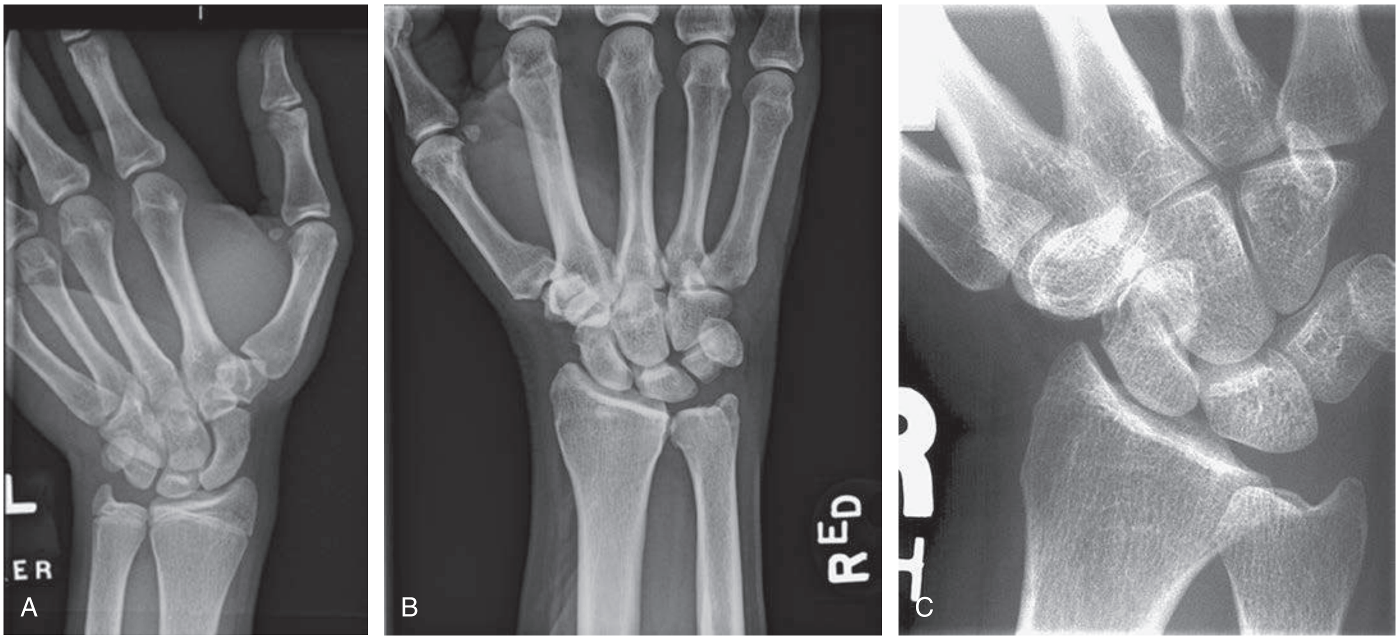


Fig. C4.162 PA wrist with ulnar deviation.

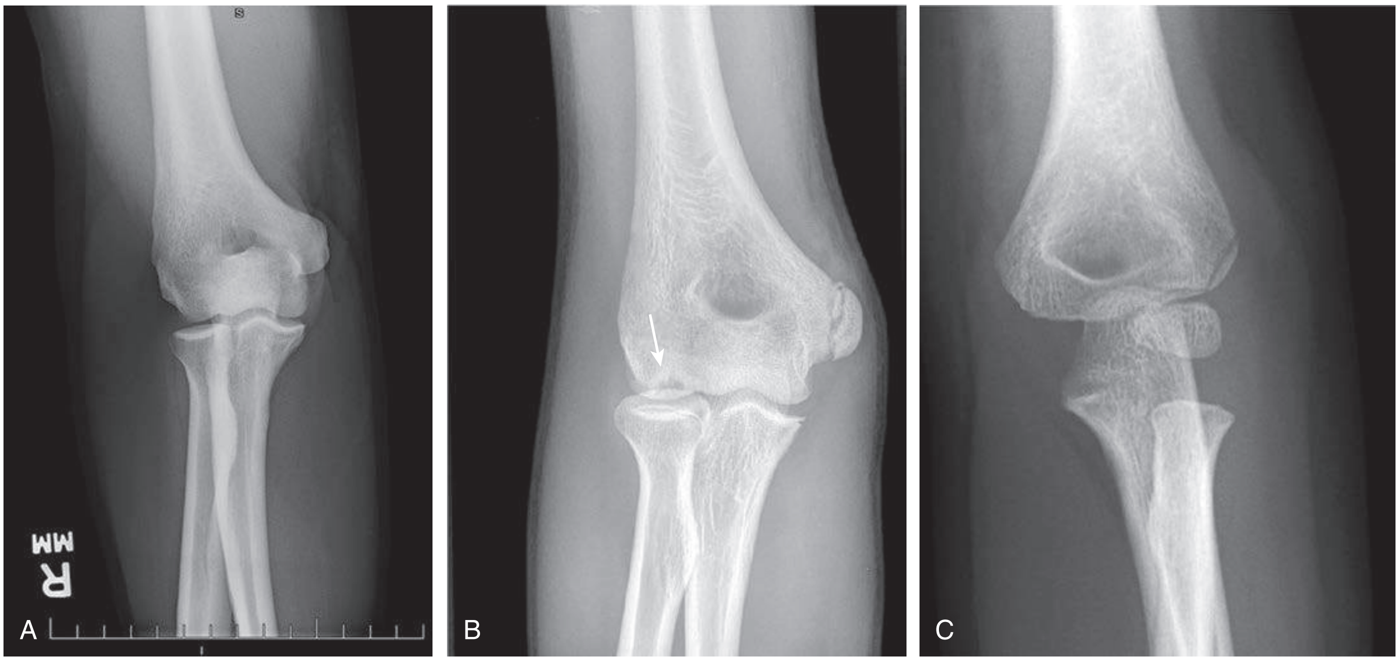


Fig. C4.163 AP elbow.

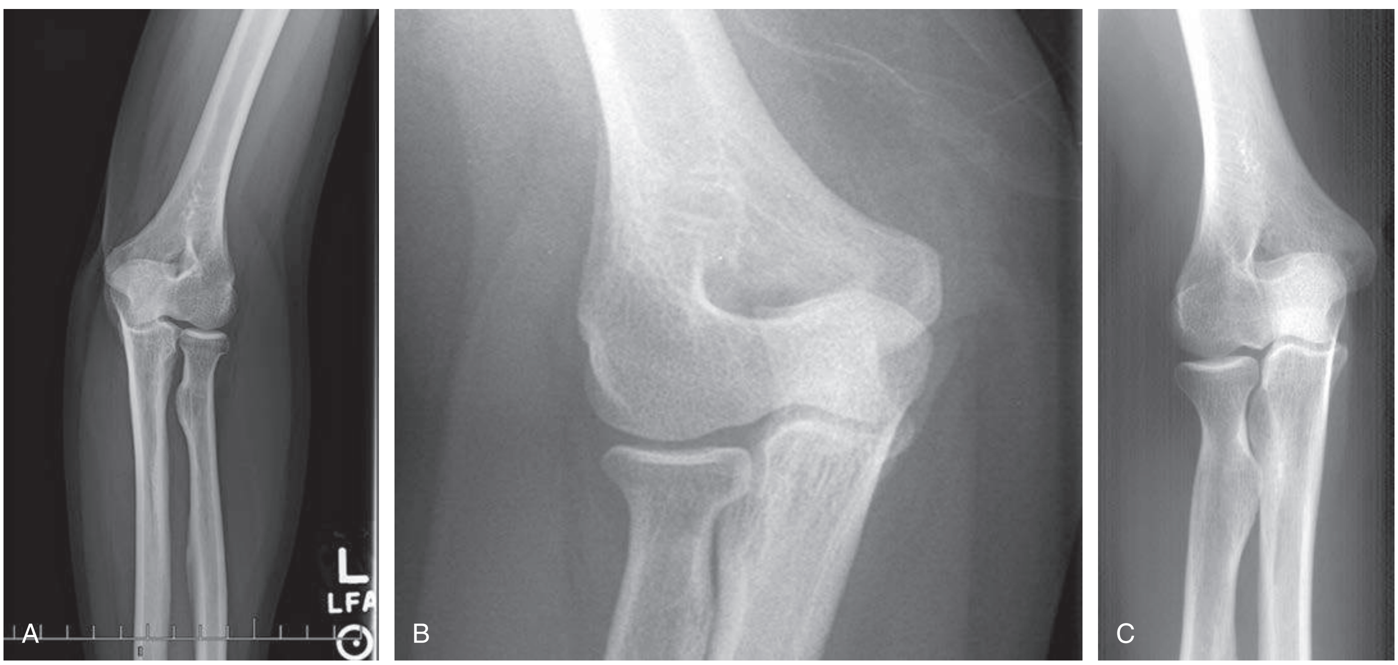


Fig. C4.164 Lateral (external) oblique elbow.

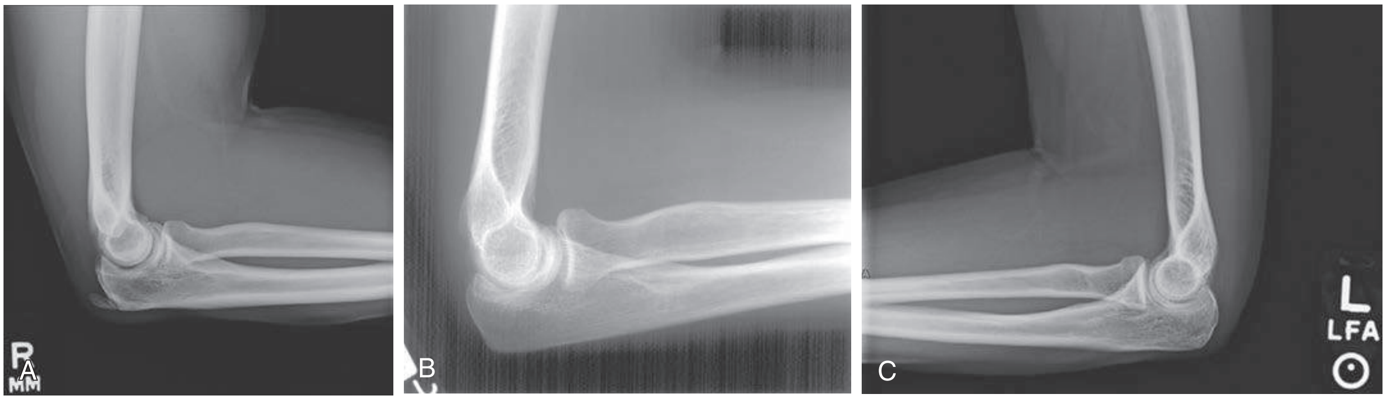


Fig. C4.165 Lateromedial elbow.